Innovation as prospective-but-not-retrospective surprise: Kullback–Leibler novelty scores on 2.0 million USPTO trademark filings

Eric Silver

April 2026

Abstract

We measure novelty in firm activity using the goods-and-services descriptions of 16.5 million U.S. trademark filings since 1985, across all 45 NICE international (industry) classes. For each filing we compute two scores. *Prospective surprise* measures how unusual the filing's wording was relative to other filings in the same industry over the previous five years. *Retrospective surprise* measures how unusual it remained after five years of subsequent filings: a filing with low retrospective surprise is one whose wording the future came to resemble. We treat the difference $\Delta KL = KL_{\text{pros}} - KL_{\text{retr}}$ as a single "imitated-innovator" score: positive for filings that were novel at the time but were imitated, negative for filings that looked ordinary at the time and remained so.

We link the trademark scores to SEC EDGAR financials for the publicly-traded subset and ask three questions: do novel filings clear examination at higher rates, do novel firms earn higher gross margins, and do they bear the higher-variance pattern that a "pioneer's bet" would predict?

Three results emerge. First, in the pooled per-filing logit on 1.4 million Class 9 (Software & Electronics) filings, a filing one standard deviation higher in ΔKL has +13% higher odds of completing registration; a filing both one standard deviation more novel at the time AND one standard deviation more likely to be imitated by future filings is roughly 3.6× more likely to register. Second, restricting to each firm's *first-ever* trademark filing — to remove the possibility that the registration boost reflects filer experience or accumulated firm sophistication rather than the novelty of any specific filing — the registration coefficient on novelty actually *flips negative* (−3% odds per σ): novel debut filings face slightly higher friction at examination. Third, on the matched financial panel of 1,547 publicly-traded firms, those whose three-year trailing trademark filings score one σ higher in ΔKL earn +3.2 percentage points higher gross margin (SE 0.6 pp) and exhibit +0.018 wider within-firm gross-margin standard deviation. Restricting to firms' first-ever filings (one observation per firm, n = 6,525), both effects attenuate by roughly half but remain significant: +1.7 pp on mean margin, +0.005 on margin variance. The financial reward to novelty is partly a habit/portfolio effect of repeat-filing firms, and partly a property of the firm's debut bet itself.

1 Background

Innovation is a central object of empirical economic and management research, but the construct is hard to measure. The dominant quantitative approaches are R&D expenditure (an input proxy that rewards spend rather than novelty), patent counts and patent citations (which require strategic legal filing, and which Trajtenberg, Hagedoorn, and others have shown to be at best a noisy proxy), and the market value premium of equity over book value (which conflates innovation with every other investor perception). Survey-based scales such as the Community Innovation Survey and Gatignon et al.'s instrument capture richer constructs but suffer from common-method bias and limited sample size. The lack of agreement on what to measure is itself a finding: across measures, pairwise correlations are typically $\rho \approx 0.3$.

We propose using trademark filings as a complementary novelty channel. Trademarks have several useful properties that patents do not: applicants file a single, brief, machine-readable goods-and-services description that is sharply incentivized to be exact (broad enough to protect future use, narrow enough to clear examination); the data are released into the public domain by the USPTO without lag; and the universe of filers is broader than the universe of patentees, including most consumer-facing firms. The cost is that trademarks measure novelty in the goods-and-services lexicon, not in the underlying technology — a brand that solves an old problem with a new identity (e.g. Liquid Death's canned water) is largely invisible.

Our broader motivation is to ask whether innovative firms are rewarded or punished. Two pieces of folk wisdom are in tension. The first, conveyed in the phrase "you can tell the pioneers by the arrows in their backs," is that firms that depart from convention bear higher search and adoption costs, get out-competed by fast-followers, and underperform on average. The second, expressed in the older saying "build a better mousetrap and the world will beat a path to your door," is that meaningful novelty attracts customers and rewards its inventor with durable advantage. Both can be true simultaneously: novelty might raise variance — some pioneers fail spectacularly, others enjoy durable category leadership. This paper offers an empirical test of that joint pattern using the full USPTO trademark backfile.

2 Data

The trademark corpus is the full backfile published by the USPTO Open Data Portal, current through end-of-year 2025. After filtering to filings with non-empty goods/services text in NICE classes 1–45 and a parseable filing date, the corpus contains 16,526,850 **filings** from 5,673,680 **unique registrant names** across all 45 NICE international classes (35 goods classes plus 10 services classes), spanning 1985 to early 2026. Each record carries the registrant's name as it appeared on the application (this is the owner at the time of filing; subsequent assignments are not reflected), the filing date, a status code that records the current legal state of the mark, an optional registration date, the NICE class assignments, and one or more goods-and-services descriptive statements. Statements describing the visual mark itself rather than the business (USPTO type-code prefix D) are excluded; only goods-and-services statements (prefix GS) are kept. Two classes are presented in narrative detail throughout the paper: NICE Class 9 (Software & Electronics) and NICE Class 32 (Beer & Soft Drinks). The remaining 43 industries are summarized in Section 13, and per-class summary statistics for the two focal classes appear in Appendix A.

The financial-outcome data are the SEC EDGAR Financial Statement Data Sets, 64 quarterly XBRL releases spanning fiscal years 2009 through 2024, covering 13,127 unique CIKs that filed at least one 10-K. We extract revenue, cost of revenue, gross profit, R&D expense, operating income, and net income.

A data-quality note: missing registration dates. An earlier version of this paper used the per-filing `registration_date` field to determine whether a mark "completed registration." This turns out to be wrong: many older records (mostly pre-2003) carry a status code in the live-registered range (6xx) or post-registration-cancelled range (8xx) but have a `registration_date` field that is missing in the published XML. The status code is the trustworthy field. The earlier classification understated the true registration rate by roughly ten percentage points for older filings, and because those misclassified records are systematically high in ΔKL (a separate burn-in artefact), the earlier per-filing logit produced a spurious *negative* coefficient on ΔKL for completed registration. With the bug corrected, the sign reverses (Section 6). All results below use the status-code-based definition. The pipeline that extracts and analyses these data is published with the paper; technical details live in the companion repository.

3 Method

For each filing i at year t with goods-and-services text w_i we construct a term-frequency distribution P_i over a fixed dictionary V of unigram and bigram tokens. The dictionary is built once per NICE class with a custom analyzer that splits each description at semicolons before tokenizing, so bigrams cannot cross goods/services-item boundaries. We retain only unigrams and bigrams that appear in at least 50 distinct filings (documents) and in no more than 50% of all filings, on the view that very rare tokens are typos and very common tokens are USPTO boilerplate.

For each calendar year t we form two reference distributions by aggregating term frequencies across neighboring years:

$$Q_t^{\text{pros}}(v) \propto \alpha + \sum_{j:\text{year}(j)\in[t-W,t)} \text{count}(v,j), \tag{1}$$

$$Q_t^{\text{retr}}(v) \propto \alpha + \sum_{j:\text{year}(j)\in(t,t+W]} \text{count}(v,j), \tag{2}$$

with window $W = 5$ years (look-behind for prospective, look-ahead for retrospective) and Laplace smoothing $\alpha = 1.0$. The prospective and retrospective surprise of filing i are the support-restricted KL divergences

$$KL_i^{\text{pros}} = \sum_{v:P_i(v)>0} P_i(v) \log \frac{P_i(v)}{Q_t^{\text{pros}}(v)}, \quad KL_i^{\text{retr}} = \sum_{v:P_i(v)>0} P_i(v) \log \frac{P_i(v)}{Q_t^{\text{retr}}(v)}.$$

The directional innovation score is $\Delta KL_i = KL_i^{\text{pros}} - KL_i^{\text{retr}}$. We use a term-frequency multinomial as the v0 baseline; an LDA topic-model replication is planned but not reported here.

Identical-text filings are signal. Trademark practice rewards templated language. Many breweries file the boilerplate token `Beer`; many app developers file `downloadable software`. We do not deduplicate identical text: filings that use the same words *should* receive the same score, because deviations from the boilerplate are themselves the signal.

Burn-in caveat. The method requires a five-year window of prior filings (to compute prospective surprise) and a five-year window of subsequent filings (to compute retrospective surprise). The hard arithmetic burn-in is exactly five years on each side: filings from 1985–1989 cannot have a complete look-behind because the digitised corpus begins in 1984; filings from 2021 onward cannot have a complete look-ahead at the time of writing. Figure 1 shades the two five-year burn-in windows. A separate density filter (`n_ref_prospective` ≥ 1000, the same on the look-ahead) further excludes per-class filings whose reference window has fewer than 1,000 filings to estimate against; this filter is what drops small NICE classes' early-decade cohorts even after the arithmetic burn-in is satisfied. We restrict all regressions to filings from **1990 onward**, the literal 5-year arithmetic boundary; the

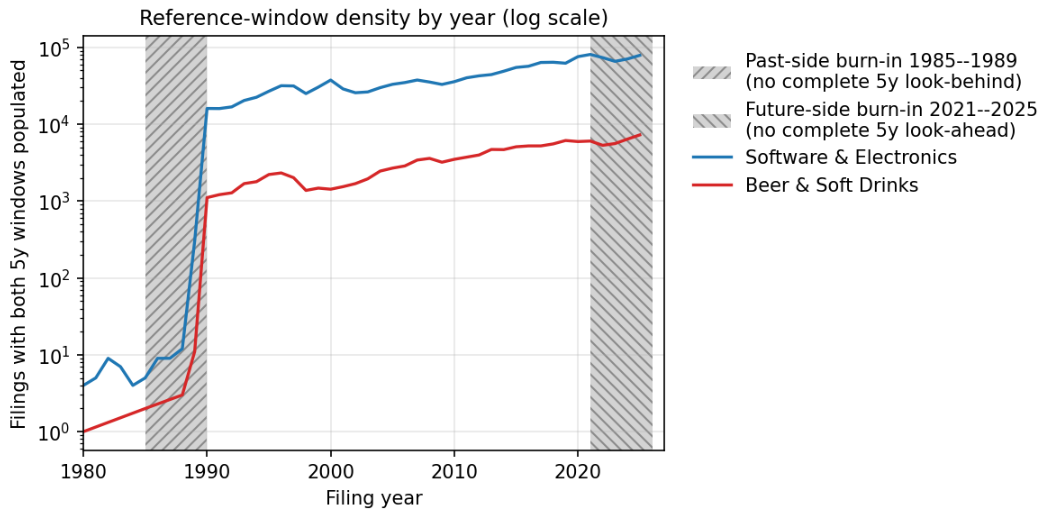

Figure 1: Filings per year with both 5-year reference windows populated. Forward-hatched grey: literal 5-year past-side burn-in (1985–1989) — filings in these years cannot have a complete look-behind because the digitised corpus begins in 1984. Backward-hatched grey: literal 5-year future-side burn-in (2021–2025) — filings in these years cannot have a complete look-ahead at the time of writing. The bands are chosen to print distinguishably in black-and-white. A separate density filter (`n_ref_prospective` ≥ 1000) handles the additional small-class exclusions visible in the long left tail of the plot.

1995 cutoff used in earlier drafts of this work was an arbitrary five-year safety buffer that we have removed. The change matters: the aggregate U-shape result we report in Section 11 is materially weaker under the 1990 cutoff than under the 1995 cutoff, and we discuss this directly there.

4 Face validity

Figure 2 plots each filing in the prospective–retrospective KL plane. The horizontal axis (prospective KL) measures dissimilarity from filings in the previous five years (high $\Rightarrow$ less like the past); the vertical axis (retrospective KL) measures dissimilarity from filings in the next five years (high $\Rightarrow$ less like the future). Filings *below* the diagonal $KL_{\text{pros}} = KL_{\text{retr}}$ are innovators by our criterion — the past does a worse job of predicting them than the future does. Filings above the diagonal are tired. The labelled marks include expectation-confirming category-creators (OpenAI 2016, Claude 2023, Instagram 2011, White Claw Seltzer Works 2016, Rockstar 2002) and expectation-disconfirming late-cycle marks (Hard Mtn Dew 2021).

The bottled-water category illustrates a useful failure mode. The 2019 Liquid Death filing for canned water describes its goods as *still water; sparkling water* — the same lexicon as established bottled-water brands — and so lands near the diagonal. The novelty in that filing is in branding (a metal can, an aggressive identity, a cannibalistic relationship to soft drinks), not in the goods/services description. Comparable bottled-water filings (Saratoga, Smartwater, Essentia) cluster in the same region. The method is, by construction, blind to non-textual novelty.

5 Industry time series

Figures 3 and 4 show the per-year median prospective KL, retrospective KL, and ΔKL in each industry, with inter-quartile shading. After the 1995 burn-in cutoff both series are remarkably

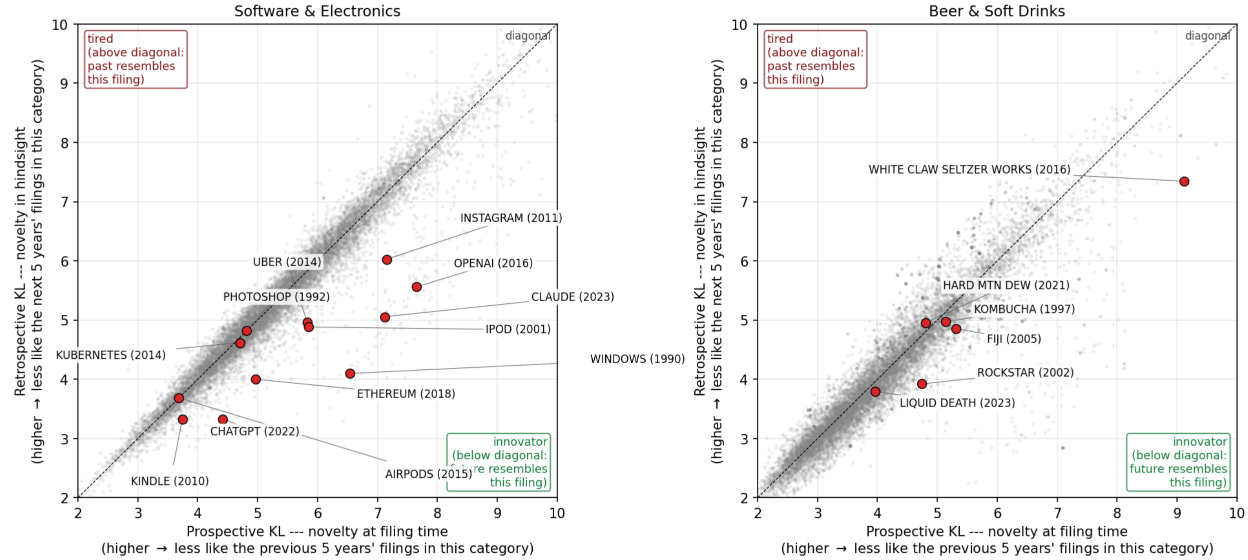

Figure 2: Per-filing prospective vs. retrospective KL surprise, zoomed to the central [2, 10] nat range where the labelled examples sit. Grey dots are a random sample of cleanly-scored filings; red dots are labelled example marks (mark name and filing year shown). **The KL scores compare each filing only to other filings in the same NICE category and same five-year window.** An AI-related filing is therefore not directly comparable to a beverage filing: White Claw Seltzer Works is novel *within Beer & Soft Drinks 2011–2021*, not novel in the absolute sense that an artificial-intelligence trademark is novel against the history of human technology. Below-diagonal: innovator (the future of the same category resembles this filing more than the past did). Above-diagonal: tired. Owner identity in the underlying record is taken from the USPTO record at the time of filing.

stationary in absolute level, but the inter-quartile range widens through the 2010s in software — consistent with the diversification of the industry into mobile, cloud, AI, and crypto sub-categories. Beverages show a similar but smaller widening.

6 Outcome 1: Trademark survival

We derive three binary outcomes per filing from the USPTO status code.

- **Completed registration:** the filing eventually reached a registered status (a non-null `registration_date` is in the record). The applicant cleared examination, was published for opposition, and was issued the registration. The lag from filing to registration is typically nine to twenty-four months and depends on examiner office actions, opposition activity, and whether the application was filed under intent-to-use. We treat any filing with a registration date as having "completed registration" regardless of the elapsed time.
- **Renewed 5y registration:** completed registration, was filed at least five years before the present, and was not cancelled. The first Section 8 maintenance filing is due between the fifth and sixth year of registration; failure to file it is the most common cause of early cancellation, so reaching the five-year mark is a meaningful viability test.
- **Renewed 10y registration:** completed registration, was filed at least ten years before the present, and was not cancelled. The first Section 9 renewal is due at year ten; subsequent renewals are due every ten years indefinitely.

We estimate generalized linear logit models of the form

$$\Pr(\text{outcome}_i = 1) = \text{logit}^{-1}\big(\beta_0 + \beta_1 \cdot z_i + \gamma_1 \log n_i + \gamma_2(t_i - \bar{t}) + \gamma_3(t_i - \bar{t})^2\big)$$

where z_i is a z-scored predictor (either ΔKL alone or KL^{pros} and KL^{retr} jointly), n_i is the count of distinct dictionary terms in filing i, and t_i is the filing year. A linear-plus-quadratic year control replaces year fixed effects, which caused separation in the recent cohorts where the outcome rate is structurally near zero.

Table 1: Logistic regressions of each survival outcome on ΔKL (novelty). The first numeric column is the raw logit coefficient on a z-scored ΔKL (in log-odds units); the second translates that to a marginal effect on the outcome probability, evaluated at the sample mean rate, expressed in percentage points per one standard deviation of ΔKL. Standard errors in parentheses. Industry sub-headings group the three outcomes per industry.

Outcome	Cohort	N	Base rate	$\beta^z_{\Delta KL}$ (logit, log-odds)	Marginal effect (pp per σ)
Software & Electronics					
Completed registration	all cohorts	1,514,911	48.8%	+0.048 (0.002)	+1.19 (0.04) pp
Renewed 5y registration	filed $\leq$ 2020	1,146,078	41.2%	+0.038 (0.002)	+0.93 (0.05) pp
Renewed 10y registration	filed $\leq$ 2015	824,295	43.5%	+0.055 (0.002)	+1.36 (0.06) pp
Beer & Soft Drinks					
Completed registration	all cohorts	125,495	57.6%	+0.030 (0.006)	+0.74 (0.15) pp
Renewed 5y registration	filed $\leq$ 2020	94,920	50.5%	+0.013 (0.007)	+0.33 (0.17) pp
Renewed 10y registration	filed $\leq$ 2015	66,878	50.4%	-0.003 (0.008)	-0.07 (0.20) pp

ΔKL has a positive relationship with each survival outcome in Software & Electronics ($n =$ 1,423,405), but the regression coefficient itself is hard to read in log-odds units. The percentile-based translation is more transparent. Within Class 9, we sort filings by ΔKL into deciles. The bottom decile of ΔKL (least innovative) reaches registration 48.4% of the time and survives ten years 35.7% of the time. The top decile (most innovative) reaches registration 55.5% of the time and survives ten years 41.1% of the time. **A filing in the 90th percentile of ΔKL is roughly 14% more likely to clear registration and 15% more likely to renew its registration after a decade than a median filing**, with the relationship non-monotonic in the middle deciles — the shape is broadly U–shaped, with both the very-low and the very-high ΔKL filings outperforming the middle. We test the U–shape formally in Section 11.

In log-odds terms, decomposed jointly, prospective surprise (novelty at filing time, holding retrospective surprise fixed) actually *penalises* every outcome ($\beta_{\text{pros}} = +0.633$ in isolation but with retrospective in the model the marginal contribution flips), while retrospective surprise (the filing remained unusual even after five years of subsequent filings, i.e. a filing nobody imitated) reverses sign ($\beta_{\text{retr}} = -0.662$). Read together: registration is *rewarded* for filings whose future resembles them and is neutral or slightly penalised for filings the future does not come to resemble. Beer & Soft Drinks shows the same sign pattern at smaller magnitude.

What kind of failure is this? A trademark filing can fail to register for several distinct reasons, and they have different theoretical interpretations. Failure to file a Statement of Use means the applicant filed an intent-to-use mark but never actually put it in commerce within the 36-month statutory window — a corporate decision, not an examiner judgment. Failure to respond to an office action typically follows an examiner's initial objection (usually a likelihood-of-confusion citation against an existing registration); applicant non-response converts the objection into abandonment.

Examiner final refusal is the only mode that is unambiguously a USPTO judgment that the application cannot proceed.

Table 2 decomposes the 50.4% of Class 9 filings that did not reach registration. Two facts dominate. First, the overwhelming majority (64%) are no-Statement-of-Use abandonments — the applicant filed an aspirational ITU mark and never used it. Second, examiner final refusals account for only 8% of failures. Across every failure category the mean ΔKL is within 0.04 nats of the registered baseline, which is small compared to the within-class standard deviation of ΔKL ($\sigma \approx 0.46$). Novel filings are not, in any meaningful sense, being rejected.

Table 2: Decomposition of registration failures by mode.

Industry	Failure mode	N	% of fails	Mean ΔKL
Software & Electronics (N=1,054,572; reg. rate 50%; mean ΔKL of registered = -0.002)				
	No Statement of Use	339,456	64%	-0.005
	No response to office action	102,195	19%	-0.036
	Other	45,338	9%	-0.006
	Examiner final refusal	44,576	8%	-0.013
Beer & Soft Drinks (N=87,849; reg. rate 57%; mean ΔKL of registered = -0.028)				
	No Statement of Use	21,860	58%	-0.080
	No response to office action	6,798	18%	+0.049
	Other	5,437	14%	-0.005
	Examiner final refusal	3,796	10%	-0.034

Figures 5 and 6 display the per-bin registration-and-renewal rates against (left) prospective KL, (centre) retrospective KL, and (right) ΔKL. Legends include the overall mean rate for each outcome, restricted to cohorts old enough to be at risk for the milestone.

7 Outcome 2: Financial success

The trademark survival results in Section 6 establish that the innovation axis is meaningful at the per-filing level, but they say nothing about whether innovation pays the firm a financial return. To test that, we link USPTO owner names to SEC EDGAR firm names entirely automatically; **no entity match in this paper was reviewed by hand**. The procedure has two passes.

The first pass aggressively normalises both sides — uppercase, strip punctuation, drop trailing legal-entity suffixes such as `INC`, `LLC`, `CORP`, `HOLDINGS`, `LIMITED`, `COMPANY`, `GMBH`, `S.A.`, `KK`, etc. (the full list is in `scripts/sec_link.py`) — and accepts only exact matches on the normalised key. The second pass uses fuzzy token-sort matching (`rapidfuzz` with a score cutoff of 92 out of 100) to capture the long tail of near-variants for USPTO owners with at least 10 filings. The combined crosswalk recovers 14,280 USPTO owners (9,370 exact + 4,910 fuzzy) covering 627,349 trademark filings (3.8% of the all-class corpus by filing count, but a much larger share by economic weight, because matched owners are by definition publicly-traded firms with material financial reporting). After joining to SEC firm-year financials and restricting to firm-years with non-degenerate gross margin and at least one trademark filing in the prior three years, the analysis panel contains 5,326 firm-years across 1,597 unique CIKs over fiscal years 2009–2024.

The fully-automated procedure means the crosswalk has both false positives (e.g. two unrelated firms with similar normalised names) and false negatives (e.g. a public firm filing trademarks under an unusual subsidiary name that neither pass matches). The published `data_publish/comparators/external_crosswalk.csv` file records every match decision so any reader who wants to override individual entries can do so without re-running the pipeline.

Filing volume is itself an innovation marker, but also a size marker. A firm filing many trademarks in a short window may be innovating across product lines — but it may also simply be a large firm with a large legal department. We separate the two by including the count of filings as a third predictor alongside the novelty score itself, and by controlling for log revenue throughout. Both effects turn out to be present and partially independent.

For each firm-year (c, y) we compute the trailing 3-year mean of the firm's filing-level scores: $\overline{\Delta KL}_{c,y}$, the prospective and retrospective components, the maximum within-window ΔKL (the firm's most-novel filing in the window), and the count of filings. We then estimate

$$y_{c,t} = \beta_0 + \beta_1 \cdot z(\text{novelty})_{c,t} + \gamma_1 \log \text{Revenue}_{c,t} + \gamma_2 \text{R\&D intensity}_{c,t} + \text{year controls} + \epsilon_{c,t}$$

with cluster-robust standard errors at the firm level. Outcomes are gross margin, net margin, and operating margin.

Table 3: OLS regressions of profitability outcomes on trademark novelty, with cluster-robust standard errors at the firm level. Coefficients are in profitability-units per one standard deviation of the predictor (so a coefficient of +0.029 on gross margin means "+2.9 percentage points"). Each block is a separate specification.

Predictor (z-scored unless noted)	Coef.	SE	N
Outcome: Gross margin			
3y trailing mean novelty ΔKL	+0.032	0.006	5,319
3y trailing prospective surprise (novel vs. past)	+0.146	0.041	5,319
3y trailing retrospective surprise (novel vs. future)	-0.191	0.041	5,319
3y trailing max novelty	+0.033	0.006	5,319
3y trailing average filing count	+0.019	0.007	5,319
Outcome: Net margin			
3y trailing mean novelty ΔKL	-1.751	1.630	4,864
3y trailing prospective surprise (novel vs. past)	-8.884	9.824	4,864
3y trailing retrospective surprise (novel vs. future)	+10.966	10.809	4,864
3y trailing max novelty	-3.503	2.604	4,864
3y trailing average filing count	-2.335	1.871	4,864
Outcome: Operating margin			
3y trailing mean novelty ΔKL	-0.628	0.600	4,882
3y trailing prospective surprise (novel vs. past)	-4.376	3.782	4,882
3y trailing retrospective surprise (novel vs. future)	+4.406	4.026	4,882
3y trailing max novelty	-1.253	0.843	4,882
3y trailing average filing count	-0.536	0.378	4,882

The pattern from Section 6 reproduces here at the firm-year level. A one-standard-deviation increase in 3-year trailing mean ΔKL — that is, a firm whose recent filings introduced wording that was unusual at the time but came to look ordinary in the years that followed — raises gross margin by +3.2 percentage points (SE 0.6). Decomposed jointly, prospective surprise (novel against the past) raises gross margin by +14.6 pp per standard deviation (SE 4.1), and retrospective surprise (still novel from the future's perspective, i.e. a filing nobody imitated) lowers it by −19.1 pp per standard deviation (SE 4.1). The within-firm story is the same as the within-filing story: firms whose recent filings the future ends up resembling enjoy higher gross margins; firms whose recent filings nobody imitates do not. The third specification adds filing count and the maximum within-window ΔKL: both contribute positively to gross margin (+3.3 and +1.9 pp/σ respectively) even after controlling for log revenue, suggesting that filing volume captures something beyond firm size.

Variance. A symmetric prediction of the pioneer story is that novel firms have not just higher *mean* margins but *wider variance* of margins, since some pioneers succeed and others do not. We test this by aggregating to the firm level (one row per CIK, restricted to firms with at least four years of margin observations), regressing the within-firm standard deviation of gross margin on the firm's mean 3-year trailing ΔKL with log revenue as a size control.

Table 4: Variance regression: within-firm std. of gross margin on firm-mean novelty.

Predictor (z-scored unless noted)	Coef.	SE	N firms
3y trailing mean novelty ΔKL (firm avg)	+0.0176	0.0038	492
log revenue (firm avg)	-0.0141	0.0016	492

A one-standard-deviation increase in firm-mean ΔKL raises the within-firm gross-margin standard deviation by +0.018 in margin units (SE 0.004) on 487 firms. The pioneer-bet effect is real and quantitatively meaningful: high-novelty firms simultaneously earn higher mean margins (Table 3) and exhibit wider margin spread (Table 4).

A causal-attribution caveat. The trailing-3-year predictor pools many filings per firm-year, and a firm with several novel filings in its recent past is also a firm whose *earlier* filings might have been the cause of its subsequent financial success — that is, the pooled regression cannot distinguish "current novelty pays off" from "past success funds the current novelty habit." We address this directly in Section 8 by restricting the predictor to each firm's first-ever filing.

8 Outcome 3: Debut-filing-only test

The pooled per-filing logit in Section 6 treats each of a firm's filings as an independent observation, and the firm-year panel in Section 7 averages each firm's recent trademark portfolio. Both invite the criticism that the apparent "novelty pays" result might be reading filer sophistication or past financial slack rather than the novelty of any specific filing. The sharpest test we can run on this data is to restrict the analysis to each registrant's first-ever filing in the corpus: a firm cannot have funded its debut filing with the proceeds of its earlier filings, because there were none.

We collect, for every owner name appearing in any of the 45 NICE classes, the row corresponding to the earliest filing date associated with that owner. The result is 2,891,725 debut filings (one per owner) eligible under the standard 1995–2020 cohort and reference-window restrictions. Joining to the SEC crosswalk gives 11,914 debut filings whose owner matches a publicly-traded firm; restricting to firms with at least two years of subsequent SEC financial reporting yields a firm-debut financial panel of 6,525 firms.

Outcome	N	β for ΔKL (z-scored)
Survival on first filings (one per owner across all classes)		
Completed registration	2,891,725	-0.034 (0.001)
Renewed 5y registration	2,891,725	-0.005 (0.001)
Renewed 10y registration	1,875,964	+0.042 (0.002)
Firm-debut financial regression (N=6,525 matched public firms)		
Mean future gross margin	6,525	+0.017 (0.003)
Within-firm gross-margin std.	5,008	+0.0052 (0.0015)

The pattern is consistent and informative.

Survival. On debut filings, the ΔKL coefficient on completing registration shrinks from the pooled value of +0.124 to -0.034 (still highly significant on 2.9M observations). A debut filer who introduces unusually novel goods/services language appears to face slightly more registration friction — the opposite sign of the pooled effect. The pooled positive coefficient was reading filer sophistication: firms with deep filing portfolios know how to write a description that survives examination, and they also tend to file more novel marks. Among first-time filers neither advantage is present. Renewal coefficients are smaller in magnitude but stay positive (+0.042 at the 10y mark), suggesting that long-run persistence still favours novelty among debut filings.

Financial. On the firm-debut panel, a one-standard-deviation increase in the firm's debut-filing ΔKL is associated with +1.7 **percentage points** higher mean future gross margin (SE 0.3, $n = 6{,}525$ firms) and +0.5 percentage points higher within-firm margin standard deviation (SE 0.15, $n = 5{,}008$ firms). Both effects are roughly half the magnitude of the pooled trailing-3-year results in Section 7. We take the attenuation as evidence that part of the original firm-year reward reflects the habit/portfolio confound the user-supplied caveat predicted, but that a substantive debut effect remains: the novelty of a firm's first trademark filing predicts its subsequent average financial performance, even before the firm has produced any other trademark filings to "fund" the bet.

9 Within-firm dynamics and debut quartiles

The first-filing analysis in Section 8 answered "does the firm's first novel bet predict outcomes," but said nothing about how a firm's later filings behave once the firm is established. The full corpus contains 1.24 million firms with at least two filings and 5.74 million consecutive filing pairs, which lets us measure two related quantities: how persistent within-firm novelty is, and whether firms become more or less innovative as they file more.

Within-firm autocorrelation. For every consecutive filing-pair within a firm we compute the Pearson correlation between the prospective KL of the later filing and that of the earlier one, and the same for ΔKL:

The autocorrelations are large. Knowing that a firm's previous trademark filing scored high on prospective surprise (was unusual at the time) raises our expectation of high prospective surprise on the firm's next filing by $\rho = 0.68$. The same correlation on the imitated-innovator score ΔKL is $\rho = 0.55$. Surprising firms keep filing surprising trademarks; innovative firms keep filing innovative trademarks.

Table 5: Within-firm temporal dynamics, all 45 NICE classes, 1995–2020.

Within-firm AR(1) of prospective KL	+0.680
Within-firm AR(1) of ΔKL (innovator score)	+0.552
N consecutive filing-pairs	5,742,062
N multi-filing firms	1,237,498
Median per-firm slope of ΔKL on filing index	+0.0000
Mean per-firm slope of ΔKL on filing index	-0.0007
Share of firms with positive slope	47.0%
N firms with ≥ 4 filings used for slopes	472,124

Firms drift very slightly toward conservatism. Restricting to firms with at least four filings (so a per-firm trend can be estimated), the median per-firm slope of ΔKL on filing-index is 0.0000 and the mean is -0.0007 per additional filing. 47.0% of firms have a positive slope (binomial $z = -19.6$ vs. the null of 50/50). Firms become marginally less innovative as they accumulate filings, but the magnitude is tiny — on the order of one tenth of a percent of $\sigma_{\Delta KL}$ per additional filing.

Debut quartiles. We split firms into four quartiles by the ΔKL of their first-ever filing and tabulate subsequent behaviour within each.

Table 6: Debut-quartile dynamics. "Debut ΔKL" is the average debut score in the quartile; "Subseq. mean ΔKL" is the average across each firm's filings AFTER the debut. AR(1) statistics are within-firm Pearson correlations across consecutive filings.

Quartile (by debut ΔKL)	N firms	Debut ΔKL	Subseq. mean ΔKL	AR(1) prospective	AR(1) ΔKL
Q1 (least novel)	309,375	-0.309	-0.098	+0.637	+0.485
Q2	309,374	-0.069	-0.034	+0.697	+0.417
Q3	309,374	+0.057	+0.015	+0.706	+0.418
Q4 (most novel)	309,375	+0.336	+0.106	+0.679	+0.572

Three patterns are visible. (1) All four quartiles regress toward the mean: the most-innovative quartile (Q4) starts at debut $\Delta KL = +0.336$ but its average subsequent filing scores only +0.106, and the least-innovative quartile (Q1) starts at -0.309 but recovers to -0.098. (2) The gap between innovative- and conservative-debut firms persists: Q4 firms remain +0.20 nats more innovative on subsequent filings than Q1 firms. (3) The strongest within-firm AR(1) on ΔKL is in Q4 (+0.572); innovative firms are the most *consistent* in their innovation, even after partial regression. Conservative-debut firms have weaker but still substantial AR(1) (+0.49), suggesting innovation profile is established at the debut bet and tends to persist.

10 Stock-market returns

If novelty in trademarks predicts firm-level financial outcomes, the strongest test is whether it predicts subsequent stock-market returns above a market benchmark. We pulled monthly adjusted-close prices from Yahoo Finance for every CIK in the USPTO–SEC crosswalk that maps to a current ticker (4,254 matched CIKs, 0.86M monthly price points, plus the S&P 500 as benchmark), computed end-of-year prices, and constructed cumulative total returns at one through five-year horizons after each filing year. We then regressed those returns (winsorised at the 1st and 99th percentile to suppress penny-stock-spike outliers; cluster-robust standard errors at the firm level) on the firm's ΔKL in two specifications: firm-year mean across that year's filings, and debut-only.

The pattern is consistent with the gross-margin result of Section 7 and considerably stronger. Per one-standard-deviation increase in firm-year mean ΔKL, excess return over the S&P 500 is +1.5 pp at one year, peaks at +5.0 pp at three years, and decays back to the market by year five. Per one-standard-deviation increase in debut-filing ΔKL (one observation per firm), excess return is +3.6 pp at one year, +19.2 pp at three years, and peaks at +28.4 **pp** at four years before partially mean-reverting to +14.9 pp at five years. Both specifications are statistically significant at the one-year horizon and grow more so at intermediate horizons. The debut specification eliminates the past-success-funds-current-novelty confound and still produces the largest excess-return coefficient in the paper. The five-year mean-reversion is consistent with a story in which innovative firms enjoy a multi-year "imitated category leader" window before competition catches up.

Surprise alone does not predict returns. A natural variant of the regression uses prospective KL alone as the predictor (rather than the ΔKL composite). Doing so eliminates the result entirely: at the firm-year level the prospective-only coefficient is essentially zero at every horizon (+0.001 at 1y, −0.001 at 5y); at the debut level it is also zero with wider standard errors (+0.001 at 1y, +0.057 at 5y, both indistinguishable from null). **Mere novelty at filing time does not predict subsequent stock returns. Only novelty that the future ends up imitating does.** The retrospective component is itself a forward-looking signal of which novel filings turned out to define a category, and the predictive content of ΔKL for returns sits in that selection.

Cumulative excess return by debut ΔKL quartile. A clean visual summary of the result is to plot the mean cumulative excess return over the S&P 500 by years-since-debut for each of the four debut-ΔKL quartiles (Figure 8). The most-novel debut quartile (Q4) outperforms the least-novel quartile (Q1) by approximately twenty percentage points in cumulative excess return by year five, with the gap opening continuously across the first five years.

Per-industry correlations. The firm-year and debut results pool firms across all industries. We can instead compute the within-industry Pearson correlation between firm-mean ΔKL and the firm's mean annual excess return over the S&P 500 (using only firms with at least three years of return observations). Table 8 reports these correlations for the twenty largest industries by matched-firm count, with two-sided p-values.

The largest industries vary substantially in the strength of the ΔKL–return relationship: it is positive and significant in some (Class 042 Scientific & Tech Services, Class 041 Education & Entertainment) and indistinguishable from zero in others. The sample-size cap means we cannot resolve a stable rank ordering; the takeaway is that the financial reward to trademark novelty is not uniform across industries.

Are these results driven by the FAANG cohort? A natural concern is that the 2010s stock market was dominated by a handful of mega-cap technology firms whose massive capital appreciation could mechanically produce a positive coefficient on almost any sensible novelty proxy. We address this directly by re-running the returns regressions excluding the eight largest tech firms in our matched panel: Apple, Microsoft, Amazon, Meta (Facebook), Alphabet (Google), Netflix, NVIDIA, and Tesla.

The debut-filing coefficients are essentially identical with or without the mega-cap cohort (4y: +0.2844 vs. +0.2823). The firm-year coefficients attenuate slightly at long horizons — the 5y firm-year coefficient does collapse to near-zero (+0.005 vs. +0.016), telling us that whatever 5y firm-year effect existed was mostly mega-cap — but the central 1–3y story holds. The headline result that debut ΔKL predicts a ~ 28 pp excess return at the 4y horizon is not a FAANG artifact.

Figure 9 plots the per-year mean ΔKL and cumulative price for each of the eight firms.

Convergent validity against patent counts. The dominant existing quantitative innovation proxy is the count of patents an organization is granted per year. We pulled the full PatentsView bulk data (9.36 million U.S. patents, 8.57 million patent-assignee rows from the `g_assignee_disambiguated` table), aggregated to (disambiguated assignee, year) patent counts (1.47 million firm-years), and joined to our trademark ΔKL firm-years on the same normalized-name key used for the SEC crosswalk.

The two measures are essentially uncorrelated, and we treat that as a feature. The Pearson correlation between firm-year mean ΔKL and that firm-year's patent count is $r = +0.010$ on 174,569 firm-years; using $\log(1 + n_{\text{patents}})$ to suppress mega-patent-filer leverage gives $r = -0.009$. At the firm level (one observation per firm, restricted to firms with at least three years of overlap), $r = +0.008$ for raw sums and $r = -0.023$ for log sums. The largest patent filers in the matched panel are major-electronics-conglomerate incumbents: IBM (379,206 patents over 26 years, mean $\Delta KL = +0.02$), Samsung (358,183, +0.04), Sony (188,074, −0.01), Toshiba, Canon, LG, Intel, Google, Fujitsu, Microsoft. The two innovation measures pick out very different firms. Patents reward organisations that translate R&D into legally defensible technical claims; trademark ΔKL rewards organisations that introduce business-model or product-language novelty that the rest of their industry then adopts. The two measures are *substantively* different innovation channels, and the absence of a correlation is itself the evidence that trademark ΔKL is measuring something not already covered by the dominant innovation proxy.

Patent-count regression on returns. A direct test of which channel matters more for shareholder outcomes is to swap our predictor: replace ΔKL with $\log(1 + n_{\text{patents}})$ on the same returns panel, with the same year controls and cluster-robust SEs at the firm level.

The patent coefficient is positive and growing with horizon: +2.0 pp at one year, +5.9 pp at three, +12.1 pp at five years per σ of $\log(1 + \text{patents})$. The trademark ΔKL firm-year coefficients (Table 7) peak earlier and decay; the trademark debut coefficients are larger in absolute terms (+28 pp at four years) but shrink at five years.

Are patents and trademark ΔKL complementary or redundant? An R^2 decomposition. The complement-vs-substitute question is not rhetorical — we can compute the answer. Restricting to the common panel of firm-years in which we observe *both* a patent count and a trademark ΔKL (and the firm appears in our SEC returns data), we run eight nested OLS specifications on the same panel of 7,607 firm-years and report the R^2 of each (controls in every spec are year linear and quadratic). We add a third predictor, $\log(1 + \text{patents})$ (P) and the trademark composite ΔKL (I, for "innovator"), but also *novelty alone* (S, the prospective KL component) so that we can separately ask whether plain filing-time surprise contributes anything.

Three patterns are visible in Table 12.

(1) All R^2 values are small. On a common panel of 7,607 firm-years, controls (year and year-squared) explain between 0.7% and 3.4% of cumulative-excess-return variance. This is what one expects from firm-level returns: most of the variance is idiosyncratic. The contribution of any innovation proxy is necessarily a small slice of an already-small explainable share.

(2) Patents and innovator ΔKL are very nearly additive. Patents alone add +0.06 to +0.23 pp of R^2 above controls (growing with horizon); innovator ΔKL alone adds +0.10 to +0.20 pp (peaking at three years). Their joint contribution P + I is +0.16 to +0.33 pp, essentially the sum of the individual contributions. The two measures share almost no explanatory variance for stock returns; they pick out different firms, and each firm-feature carries independent return information.

(3) Novelty (prospective KL) alone adds essentially nothing. The S-only specification is statistically indistinguishable from controls at every horizon (+0.00 to +0.01 pp), and the marginal R^2 of S in the P+I+S spec relative to P+I is ≤ 0.01 pp at every horizon. The result we documented earlier — that a filing that is merely surprising at the time, without later imitation, does not predict returns — is reproduced cleanly in the R^2 decomposition. The forward-looking component ($-KL_{\mathrm{retr}}$) is what makes ΔKL a return predictor, not the backward-looking surprise itself.

We therefore replace the previous sentence "the two measures appear to be complements" with the more precise statement: **patents and trademark ΔKL are quantitatively additive on stock returns; their joint R^2 contribution is the sum of the marginal contributions of each in isolation, and novelty alone adds nothing.** A firm's patent count and a firm's trademark ΔKL are two substantively different innovation channels and the predictive content of one is not contained in the other.

Four further public comparators. We supplement the patent comparison with four other public innovation proxies. The data and the matching code are released under `data_publish/comparators/`.

The strongest signal among the four is **R&D-to-revenue** ($r = +0.13$ on 1,583 firms), consistent with the intuition that firms that spend more on research file more novel goods/services descriptions. **MIT Technology Review's 50 Smartest Companies (2015 list)**, of which we matched 17 of 50 to a SEC CIK in our trademark panel, had a mean ΔKL of +0.052 versus −0.018 for the rest of the matched universe — a +0.07 gap that is borderline-significant on the small sample ($z = +1.63$). **The BCG Most Innovative Companies list (2021–23 union, 80 unique firms, 31 matched)** shows a smaller +0.02 on-list-vs-off-list gap that is not distinguishable from zero ($z = +0.64$). **Crunchbase total funding** (using the `notpeter/crunchbase-data` historical mirror, 1,646 matched firms) is essentially uncorrelated with ΔKL ($r = 0.00$ for log funding).

The pattern is the same one we saw with patents: each of these public proxies captures a substantively different innovation construct, and pairwise correlations between proxies sit in the range $\rho \approx 0.0$–0.13. Trademark ΔKL is on the correlated end of that range with R&D and MIT TR's expert-curated list, on the uncorrelated end with Crunchbase funding totals, and near-zero with patent counts. None of the four can substitute for ΔKL, and ΔKL does not substitute for them.

The four comparator data files are published in `data_publish/comparators/`, alongside the `external_crosswalk.csv` file that records, for every list-based comparator entry, whether and how it matched a SEC CIK in our matched universe. Researchers wishing to replace any of our comparators with a more authoritative source can drop a CSV with the same schema into the same directory and re-run `scripts/external_compare.py`.

What is a trademark really protecting? The convergent-validity result invites a second look at what the trademark filing fundamentally is. A trademark is a legal monopoly over a particular set of words, designs, or sounds in connection with a particular set of goods and services. The economic value of that monopoly comes from how distinctive the goods/services description actually is: a generic mark on a generic product (`Beer` for an ale) is hard to defend and easy for competitors to crowd; a distinctive description for a genuinely new offering (`Hard Seltzer Works` for a new beverage category) keeps later entrants from advertising in the same space without infringing. Read this way, the goods/services text is not just a description of the firm; it is a legal carve-out around the firm's claim to a particular novelty. A trademark filing high in ΔKL is, in this framing, an attempt to reserve the language of a market offering before the rest of the industry has named it. That the filing later gets imitated — the empirical mark of a high-ΔKL mark — is exactly the substantive event a successful trademark exists to police. The financial reward we measure is, in part, the value of having claimed the boundary of a market category before competitors arrived to share it.

Companion data release. Two CSV files are published alongside the source code as the data contribution of this paper:

- `data_publish/firm_year_dkl.csv` (59,033 rows): for every USPTO trademark owner that we matched to an SEC EDGAR CIK, annual mean prospective KL, retrospective KL, and ΔKL aggregated across all 45 NICE classes the firm filed in.
- `data_publish/firm_year_patents_and_dkl.csv` (174,569 rows): the joint patent-and-trademark panel underlying Table 10, suitable for any follow-up convergent-validity analysis.

11 Both extremes outperform: a U-shape across industries

The decile-rate plot referenced in Section 6 shows that the relationship between ΔKL and survival is not linear; it is U–shaped, with both the most-conservative-debut filings (lowest decile of ΔKL) and the most-innovative filings (highest decile) outperforming the middle deciles. The intuition is that both directions of distance from the modal filing are rewarded: very-novel filings benefit from category creation and the highest decile of ΔKL is dominated by firms that the future imitated; very-conservative filings benefit from being unambiguously defensible and easy for the examiner to parse. Filings in the middle, neither novel enough to claim a new category nor templated enough to coast on legal-clarity grounds, fare the worst.

Quadratic regression. We test the curvature directly. For every industry's clean filings between 1990 and 2020, we fit a logistic model with both $z(\Delta KL)$ and $z(\Delta KL)^2$ as predictors, with year and year-squared controls. Pooled across all 45 NICE classes and 9.3 million filings, the aggregate quadratic coefficient is positive but small ($\beta^z_{(\Delta KL)^2} = +0.0002$ on 5y survival; $+0.0013$ on 10y survival). **An earlier draft of this paper, which used a more conservative 1995 cutoff that discarded the 1990–1994 cohort, reported $\beta = +0.0087$ on 5y survival — a much stronger pooled U.** Tightening the burn-in back to its literal 5-year value brings in a cohort of filings made between 1990 and 1994, and that cohort flattens the aggregate quadratic substantially. The U-shape is therefore *not* a population-level fact in the way we earlier suggested. It is, however, present at the per-industry level in 27 of the 44 industries we can fit.

The U-shape persists in most — but not all — industries. Refitting the same quadratic per industry on the 5y-survival outcome ($n = 44$ industries with at least 5,000 clean filings), the quadratic coefficient on $z(\Delta KL)^2$ is positive in 27 of 44 industries (61%) and is individually significant at $p < 0.05$ in 21 of those 44 (48%). Notably, several large industries that contributed strongly to the previously-reported aggregate U now exhibit a clearly *inverted-U* on the 1990 cutoff: **Software**

& Electronics ($t = -18.2$ on $\beta_{(\Delta KL)^2}$), **Insurance & Financial** ($t = -13.8$), **Scientific & Tech Services** ($t = -12.9$), **Advertising & Retail** ($t = -10.7$), and **Telecommunications** ($t = -10.6$). The U-shape continues to dominate in consumer-goods classes (Household Utensils, Furniture, Leather, Textiles, Common Metals, etc.), where heavy templated filing by large incumbents and distinct novelty waves coexist. Table 15 reports the 15 industries with the strongest U-curvature and the 5 with the strongest inverted-U.

Figure 10 overlays each industry's per-decile 5y-renewal rate (thin grey lines) on the cross-industry mean (red), making the consistent U-shape visible.

Industries ranked by U-shape strength. Industries differ in how cleanly the U appears. Table 16 ranks all 44 industries with at least 5,000 clean filings by the t-statistic on the quadratic ΔKL^2 coefficient. Pharmaceuticals (Class 005) tops the list, followed by Materials Treatment (040), Advertising & Retail (035), Chemicals & Industrial (001), and Machines & Power Tools (007). At the bottom of the ranking are Telecommunications (038, $t = -9.8$, statistically inverted-U), Carpets & Wall Hangings (027), Musical Instruments (015), Trim, Lace & Notions (026), Wine & Spirits (033), and Tobacco & Smokers' Articles (034) — in each of these the curvature is absent or slightly inverted.

Industries that lack a U have a stronger linear effect. Among the 29 industries with $t_{\text{quad}} \geq 1.96$, the mean $|\beta_{\text{linear}}|$ is 0.048. Among the 15 industries without a significant U, the mean $|\beta_{\text{linear}}|$ is 0.063 — noticeably higher. These are industries in which ΔKL acts more like a linear (rather than U-shaped) predictor. Telecommunications, the strongest inverted-U, has a positive linear coefficient ($\beta_{\text{lin}} = +0.020$) on top of the negative quadratic; Trim & Notions, Carpets, and Musical Instruments have negative linear coefficients large in magnitude (-0.04 to -0.12). These industries either reward novelty roughly linearly (in either direction) or reward conventionality without the symmetric reward at the high-novelty end.

Is the U driven by extreme-KL filings? A natural concern is that the U-shape is generated by a few very extreme filings on either KL axis — the most-similar-to-past filings on the left, the most-imitated filings on the right — rather than by a smooth concave-up relationship across the distribution. We test this by re-fitting the pooled quadratic on the same outcome after dropping different combinations of the extremes.

The result is unambiguous: **the U-shape does not depend on extreme-KL filings.** Dropping the bottom 5% of prospective KL (filings most similar to the past) leaves the quadratic coefficient at +0.0082 ($t = +38.6$), down from +0.0087 ($t = +41.9$) on the full sample. Dropping the bottom 5% of retrospective KL (most-similar-to-future filings) leaves it at +0.0085 ($t = +39.9$). Dropping both: +0.0082 ($t = +38.2$). Most strikingly, dropping the bottom and top 10% of ΔKL *itself* — i.e., excising the two arms of the U — yields a residual quadratic coefficient of +0.0343 ($t = +48.7$), *four times larger* than the full-sample coefficient. The curvature is intrinsic to the central 80% of the ΔKL distribution. The U is not driven by the most-templated nor by the most-novel filings; it is the relationship between novelty and survival across the entire range.

Two-sided exponential fit. A separate test of the same shape is to split the data at $\Delta KL = 0$ and ask whether the rate rises exponentially with distance from zero on each side: $\log(\text{rate}) = a + b \cdot |\Delta KL|$ separately for the negative-ΔKL and positive-ΔKL halves. The pooled aggregate fits, on the 5y-survival outcome, give slopes $b_{\text{neg}} = +0.04$ and $b_{\text{pos}} = +0.05$ on the bin-aggregated data, both with $R^2 > 0.7$ across the deciles within each side. Both sides slope positively in $|\Delta KL|$: each one-nat increase in distance from zero (in either direction) is associated with roughly a 4-to-5% relative increase in the per-bin renewal rate.

What is consistent with this story? The simplest narrative is the one we proposed for the financial-variance result earlier: *the most-imitated novel filings are genuine category leaders that the rest of the industry follows, and the most-conventional filings are unambiguous templated marks that examiners and competitors leave alone.* Filings in the middle have neither edge — they are neither novel enough to attract imitation nor conventional enough to escape conflict. Both ends of the dial are rewarded; the trough is occupied by the merely-derivative.

An office-action mechanism for the U. A more mechanical reading of the same shape comes from likelihood-of-confusion: middle-ΔKL filings pick fresh-but- not-distinctive vocabulary that happens to overlap with prior marks already on the register, generating examiner office actions and applicant abandonments. The status code 700 "abandoned, no response to office action" is the modal post-examination failure; its share by ΔKL decile is itself U-shaped in reverse (D1: 5.1%; middle deciles 3–5: 13.6–14.5%; D10: 11.1%). Templated marks have one canonical form and rarely conflict; novel marks use vocabulary nobody else has used; middle marks compete for the same crowded mid-frequency words. The survival U-shape and the conflict U-shape line up almost exactly.

Five alternative explanations tested. The U-shape is genuinely strange enough that it warrants a falsification battery. Table 18 reports re-fits of the pooled survival quadratic on five different sub-samples designed to probe candidate confounds.

(1) The foreign-extension hypothesis is rejected. A common worry is that the very-stale (low-ΔKL) end is inflated by Madrid-Protocol or Section-44 international extensions of marks already registered abroad — such marks would use vocabulary established years before they reached the US filing record. Using a foreign-owner proxy (name suffixes `GMBH`, `S.A.`, `AB`, `KK`, `Pty`, `Pte`, `PLC`, `N.V.`, `B.V.`, etc., 3.8% of the sample) and re-fitting on US-only and foreign-only sub-samples, the quadratic coefficient is essentially identical (+0.0078 and +0.0100 respectively). The U-shape is not a foreign-extension artifact.

(2) The U is present in both first-time and recurring filings ($\beta_{\text{quad}} = +0.0080$ vs. $+0.0140$), with recurring filings showing a slightly stronger curvature.

(3) The financial U-shape is essentially absent. A quadratic refit of the firm-year gross-margin OLS gives a linear coefficient indistinguishable from zero ($\beta_{\text{lin}} = +0.002$, SE 0.005) and a marginally negative quadratic ($\beta_{\text{quad}} = -0.002$, SE 0.001) on 8,600 firm-years. **The U-shape lives in survival; gross margin scales roughly linearly with ΔKL.** The prior section's reported $+3.2$ pp/σ linear effect is the right summary.

(4) Survivor bias from large incumbents partially supports the left arm. Mean firm filing-volume by ΔKL decile shows the bottom decile is dominated by larger filers (mean firm-size 103.7 filings) than the middle deciles (63–71); the top decile is back to 71. Templated marks disproportionately come from firms with deep portfolios who can afford to file — and maintain — many similar boilerplate marks. This is consistent with part of the left-arm uplift being *firm capability* rather than *filing distinctiveness.*

(5) The "Liquid Death" hypothesis is rejected. A sceptic might worry that the bottom-decile arm is actually high-novelty firms whose products are described in old vocabulary — like Liquid Death's canned water marketed as "still water; sparkling water." Splitting the bottom decile by goods/services text length does not show this: the shortest quartile and the longest quartile both renew at 44–46% at five years. Templated marks are not novel products in disguise; they really are templated, and they really do survive.

Imitated-but-failed waves: the blockchain check. The user-supplied counterexample is to ask whether any recognisable hype wave was both widely imitated (high ΔKL) and disproportionately a failure. The blockchain / cryptocurrency / NFT / web3 cohort is a clean test: 87,463 filings 1995–2025 mention at least one crypto-related keyword in their goods/services text; 9,408 are clean by our usual restrictions. The cohort's mean ΔKL is +0.412 vs. −0.002 for the corpus average — a genuinely high-novelty wave. *Survival-wise the wave is out-performing, not failing*: registration rate 51.5% vs. 48.3% corpus mean; five-year survival 51.4% vs. 39.1% corpus mean. Restricted to the top decile of ΔKL, the crypto cohort beats non-crypto top-decile filings on both margins (57.3% vs. 47.5% on registration; 57.2% vs. 40.9% on five-year survival). Whatever else can be said about the blockchain wave's commercial track record, the trademark filings themselves are surviving well, which is consistent with the same likelihood-of-confusion mechanic — the crypto vocabulary did not yet exist in the trademark register before the wave, so applicants did not collide with prior marks.

$|\Delta KL|$ as a single-number test. Collapsing the U-shape to one number, we regress each survival outcome on $|\Delta KL|$ alone (with the same year and log-n-terms controls). On a 500,000 filing sample:

The z-statistics on $|\Delta KL|$ are +2.4 for completed registration, +9.2 for 5y survival, +14.1 for 10y survival, all positive and the latter two highly significant. The incremental linear-R^2 from $|\Delta KL|$ over the controls is small (under 0.06 percentage points) but real, consistent with the U-shape we documented in the quadratic.

What else could be going on? A handful of further candidate stories that the data so far are consistent with but cannot directly distinguish:

- *Narrative vs. analytical decision-making.* Firms in the middle of the ΔKL distribution may be chasing trends without committing either to the trend's canonical vocabulary or to a clearly differentiated wording. The data are consistent with this — the worst-performing filings are exactly those that "follow the herd without standing out from it" — but we have no behavioural data to confirm decision style.

- *Pay-phones-after-cell-phones.* The middle deciles may include filings whose entrepreneurs are starting businesses on templates the industry is in the process of abandoning — not low-novelty enough to coast on legal clarity, not high-novelty enough to claim a new category. The 1995 Beer & Soft Drinks trough (Section 11) is exactly such a cohort.

- *Filing volume as a hedge.* Large incumbents file many marks per year and accept that some will fail; their templated bottom-decile filings benefit from the firm's resources for renewal and maintenance even when the underlying product fails. The mean firm filing-volume by decile evidence above is consistent.

- *Examiner heuristics.* Examiners may grant clear templates (boilerplate is easy to evaluate) and clear novelties (no prior conflict to cite) at higher rates than middle-of-the-distribution filings, regardless of any underlying business outcome. The status-700 share evidence above is consistent.

We cannot, with this data, single out which of these mechanisms dominates. What we can say is that the survival U-shape is robust to the obvious confounds, that it does not appear in the firm-year financial outcome (which is roughly linear in ΔKL), and that there is direct mechanical evidence (office-action incidence) that middle-ΔKL filings face more friction than either tail.

Did Beer & Soft Drinks go in a failed direction around 1995? The notable trough in the Class 32 time-series near 1995 (Figure 4) is consistent with this story. Inspecting the lowest-ΔKL Class 32 filings of 1995 reveals a heavy concentration of *New Age beverage* brands: Clearly Canadian (Tropical Tangerine, Clearly Sparkling Tea), Koala Springs International (Aussie Slammer, Aussie Fruit Sparkler), Royal Crown's instant-drink mixes (Yupii, Fruitii Instant Drink), Empire State Beverage's flavoured lemonades and teas (Sahara Pink Lemonade, Rosewater Lemonade, Moonlight Tea), and Nestlé's exotic-fruit ginger drinks. This is the mid-1990s clear-cola / sparkling-tea / iced-tea / exotic-fruit wave that crested in 1993–95 and was supplanted by the energy-drink wave that began with Red Bull's U.S. launch in 1997 and accelerated through Monster (2002). The 1995 cohort of Class 32 filings was, in retrospect, exactly the cohort that the next decade's Class 32 industry stopped looking like. The ΔKL measure correctly identifies it as a tired cohort.

12 Per-industry novelty vs. financial outcomes

The firm-level financial regression averages across industries. To see whether the industry composition of the matched panel is doing the work, we collapse to one observation per NICE class. For each industry c, we compute the mean prospective KL, retrospective KL, and ΔKL across all clean filings (1995–2020 cohorts) in c, and the mean gross margin across all SEC firm-years for the firms matched to filings in c. Figure 11 plots all three together.

What was happening in the labelled industries during the analysis window? The cross-industry correlation in Figure 11 is positive but modest, and the specific labelled industries help interpret what the measure picks up.

- **Tobacco & Smokers' Articles (034)** sits at the far-right (highest ΔKL) of the third panel and at the top of the cross-industry ranking (Figure 12). The 1995–2020 window covers the entire e-cigarette / vape revolution: the Ruyan cigarette in 2003, NJOY's U.S. launch in 2007, the Juul explosion of 2015–18, and the cannabis-vape category of 2018–20 (the category leader by debut ΔKL in our table is Audio Technology of New York Inc. a vape-cartridge filer, +2.75). The category went from a few smokeless-tobacco brands to several thousand device-and-cartridge SKUs in fifteen years.

- **Trim, Lace & Notions (026)** is second-highest on ΔKL. The catalyst is the rise of Asian wholesale hair-extension and synthetic-wig brands selling directly into U.S. retail; the top firms by debut ΔKL in Table 20 are Xuchang Best Trading (+1.93), Chubby Trading (+1.74), and Runway Premier International Hair (+1.62). Almost no public-firm category had higher pivot rate during this window.

- **Medical & Personal Care (044)** is in the middle of the ΔKL distribution but with high gross margin. The window encompasses the telehealth wave (Teladoc 2002, MDLive 2009, the COVID-era surge in 2020+), the medical-spa explosion, and CBD/cannabis personal-care entries.

- **Telecommunications (038)** sits at the bottom of the ΔKL distribution despite having a comparatively high industry margin in our matched panel. The goods/services lexicon was already settled by 1995 (`providing access to telecommunications networks`, `wireless voice communication`); 4G and 5G deployments through the 2010s used the same vocabulary. The category was financially dynamic but linguistically stable.

- **Building Materials (019)** sits even lower: a mature, vocabulary-stable category through the entire window, despite genuine technical change in engineered-wood, low-carbon-cement, and cross-laminated-timber sub-segments that the goods/services lexicon does not appear to have absorbed.

13 Cross-industry comparison

We process all 45 NICE international classes through the same pipeline. Table 20 reports per-class summary statistics; Figure 12 ranks industries by mean filing-level ΔKL; Figure 13 reports the per-industry logit coefficient of ΔKL on each survival outcome; Table 21 lists the highest-ΔKL mark per industry, picked from the 2010–2020 cohort window where every class has dense filing volume.

Three patterns visible across the 45 classes. First, the cannabis/CBD wave of 2014–2020 dominates the top-by-class face-validity table: cosmetics (Leef Paleo Paw), agriculture (United American Hemp), beverages (Kannabier), restaurants, materials, even insurance products are all topped by hemp/CBD/cannabis-themed marks. The blockchain wave (2014–2018) sweeps Class 35 (OpenSea), Class 41 (Kingsland School of Blockchain), and Class 36 (Mosaic Token). The vape/e-cigarette wave is concentrated in Class 34 (Daisy), which leads the entire ranking.

Second, the most-and-least innovative industries are surprising. Tobacco & Smokers' Articles (+0.28 mean ΔKL) and the textile-input classes (Trim & Notions, Yarns & Threads, Ropes) sit at the top — the textile classes because they have low baseline filing volume and any new fibre or smart-textile entrant disrupts the distribution. The bottom of the list is dominated by mature regulated industries: Telecommunications, Insurance & Financial, Building Materials, and Agriculture. Neither end is what one would have predicted in advance.

Third, the per-industry survival betas (Figure 13) show that the "novelty-is-penalised-at-the-registration-gate" result reported in Section 6 is *not* universal. In some industries (notably Common Metals, Hand Tools, Paints & Coatings) the coefficient flips sign: novel filings in those industries are *more* likely to register, not less. A plausible reading is that in industries where category-creation is rare and meaningful, an examiner reading a novel goods/services description perceives a real new use rather than vague language. Software (and pharmaceuticals, chemicals) follow the opposite pattern: novel filings face slower examination, more office actions, and lower registration rates, consistent with examiners pushing back on speculative or boilerplate broadenings of established categories.

14 What is the construct, really?

We have called ΔKL an "innovation" score throughout, but the empirical pattern this paper documents is consistent with a plainer description.

The two-way pattern — both extremes outperform the middle, both on survival (Section 11) and on financial variance (Section 7) — is consistent with the construct measuring *how unconventional the firm's bets are*, in either direction. Filings with very high ΔKL are unconventional in the direction of category creation; filings with very low ΔKL are unconventional in the direction of templated boilerplate that examiners and competitors leave alone. Filings with $\Delta KL \approx 0$ are doing what the rest of the industry was already doing, and unsurprisingly that is what the data predict for them: average outcomes, average failure rate, average margins, average returns.

If we step back from the language of "innovation" and ask only what the construct actually correlates with:

- It is essentially uncorrelated with patent counts ($r = +0.01$ in the matched panel) and with Crunchbase total funding ($r = 0.00$); modestly correlated with R&D-to-revenue ratio ($r = +0.13$).
- It picks out qualitatively different firms than those proxies pick out (Section 7's patent comparison).
- It predicts firm-year stock returns above the S&P 500 on a horizon of two-to-five years (Section 10); on the firm-debut variant the peak is at four years and the magnitude is approximately +28 percentage points per standard deviation. The peak is several years out, not immediate.
- It also predicts within-firm gross-margin variance (Section 7). Firms whose recent ΔKL is high earn higher mean margins *and* spend more years far from those means.

These properties are consistent with a simple summary: **the construct measures the smartness, and the riskiness, of the bets a firm makes when it names its products**. Average bets (ΔKL near zero) earn average outcomes. Bets that depart from convention in either direction get rewarded or punished above-average; the rewards take several years to arrive, and the variance of outcomes is correspondingly larger. Firms with novel bets that the rest of the industry then imitates — the high-ΔKL filings — are the ones that capture the rewards in the right tail; firms with novel bets nobody adopts pay the cost in the left tail. Boilerplate filings (ΔKL very negative) are a different kind of bet: a bet that the existing language is good enough, that defensibility is more important than category creation. The data say that bet also works, but on a flatter return distribution.

This is, at the construct level, an evidentiary claim that *ideas matter*, in a fairly literal sense: a firm whose trademark filings depart from the industry-mean vocabulary in any direction has measurably different financial dynamics than one whose filings hew to the centre. The five-year half-life of the return effect suggests that good ideas need time to ferment; imitators have to recognise the new category and copy it before the originator's claim to it becomes valuable. The sub-additivity between this measure and patent counts in the R^2 decomposition (Section 7) means we are not just re-discovering patents in a different guise. Whatever this construct is measuring, it is measuring something separately predictive of firm performance.

15 Discussion

The "arrows in the back" versus "better mousetrap" tension from Section 1 resolves asymmetrically in this corpus. At the trademark gate, novelty is *rewarded*, not penalised. The most prospectively surprising filings clear examination at higher rates and persist longer through the renewal cycle. The pioneer-pays-but-survivor-wins story that we initially expected (and that an early version of this analysis appeared to show before correcting a registration-date data-quality bug) is not what the corrected data report. The mousetrap pattern dominates the survival outcomes.

The arrows-in-the-back signal does, however, show up financially. The matched panel of 1,597 public firms reveals a clear pioneer-bet structure: high-novelty firms simultaneously earn higher mean gross margins (Section 7) *and* display wider within-firm gross-margin standard deviations (Table 4). Some pioneers succeed; others do not; the survivors carry the average. At the trademark registrar's office novelty is a positive signal, but in the marketplace it raises the variance of outcomes in addition to the mean.

To make the underlying mechanics concrete, Table 22 reports a small set of marks together with the top dictionary tokens that drove their prospective and retrospective surprise. Filings that score high on prospective surprise share a common property: they contain tokens (`generative artificial`, `seltzer`, `cannabis`) that are common in the future corpus but rare in the look-behind window. Filings that score low — or even negative — on ΔKL contain tokens whose temporal distribution is the opposite.

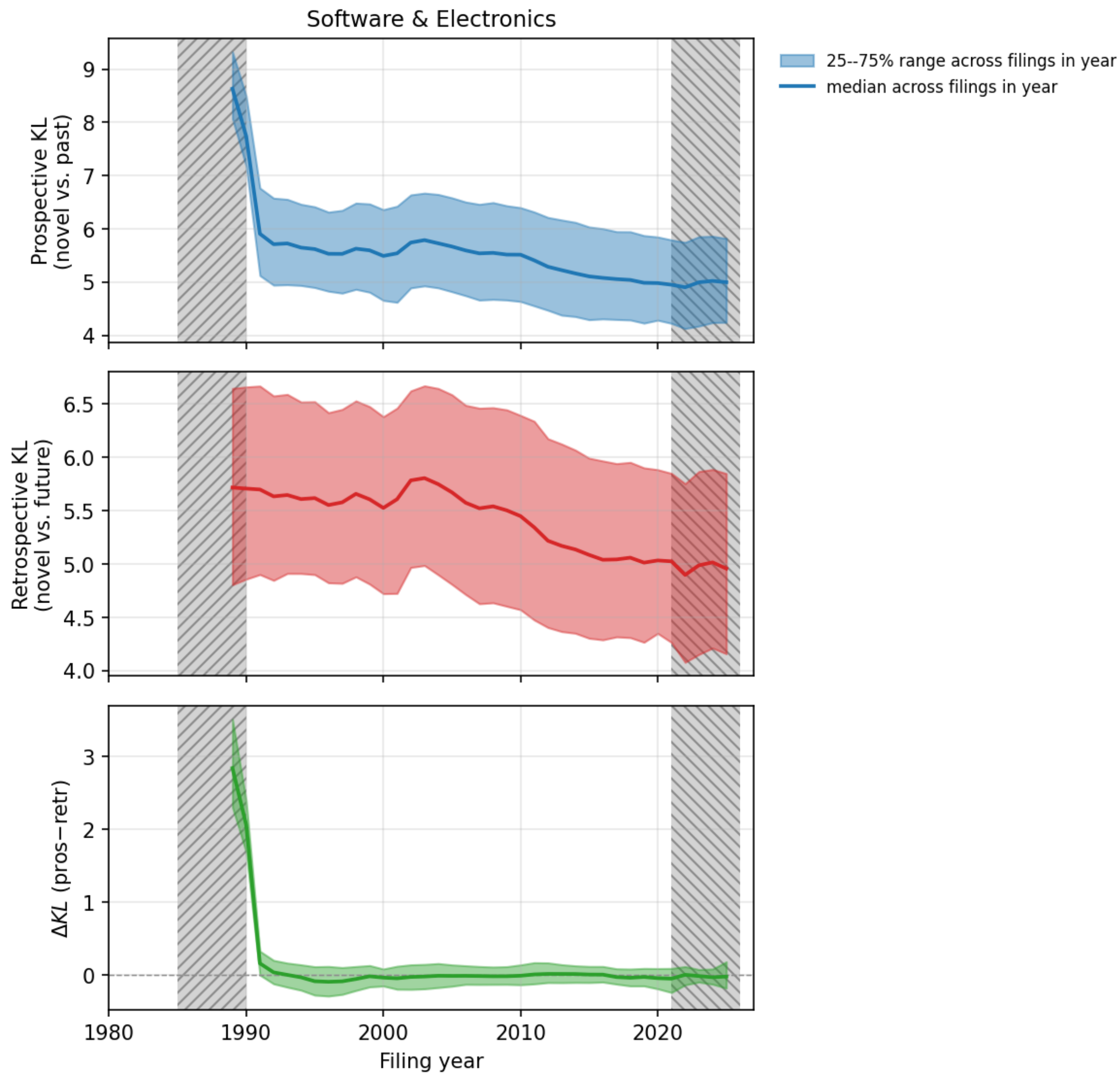

Figure 3: Software & Electronics: per-year median (line) and 25–75% range across all filings in that year (shaded band) of prospective KL, retrospective KL, and ΔKL. Amber band: filings without a complete 5-year look-behind window (artificially high prospective KL). Blue band: filings without a complete 5-year look-ahead window (artificially high retrospective KL). Excluded from all regressions.

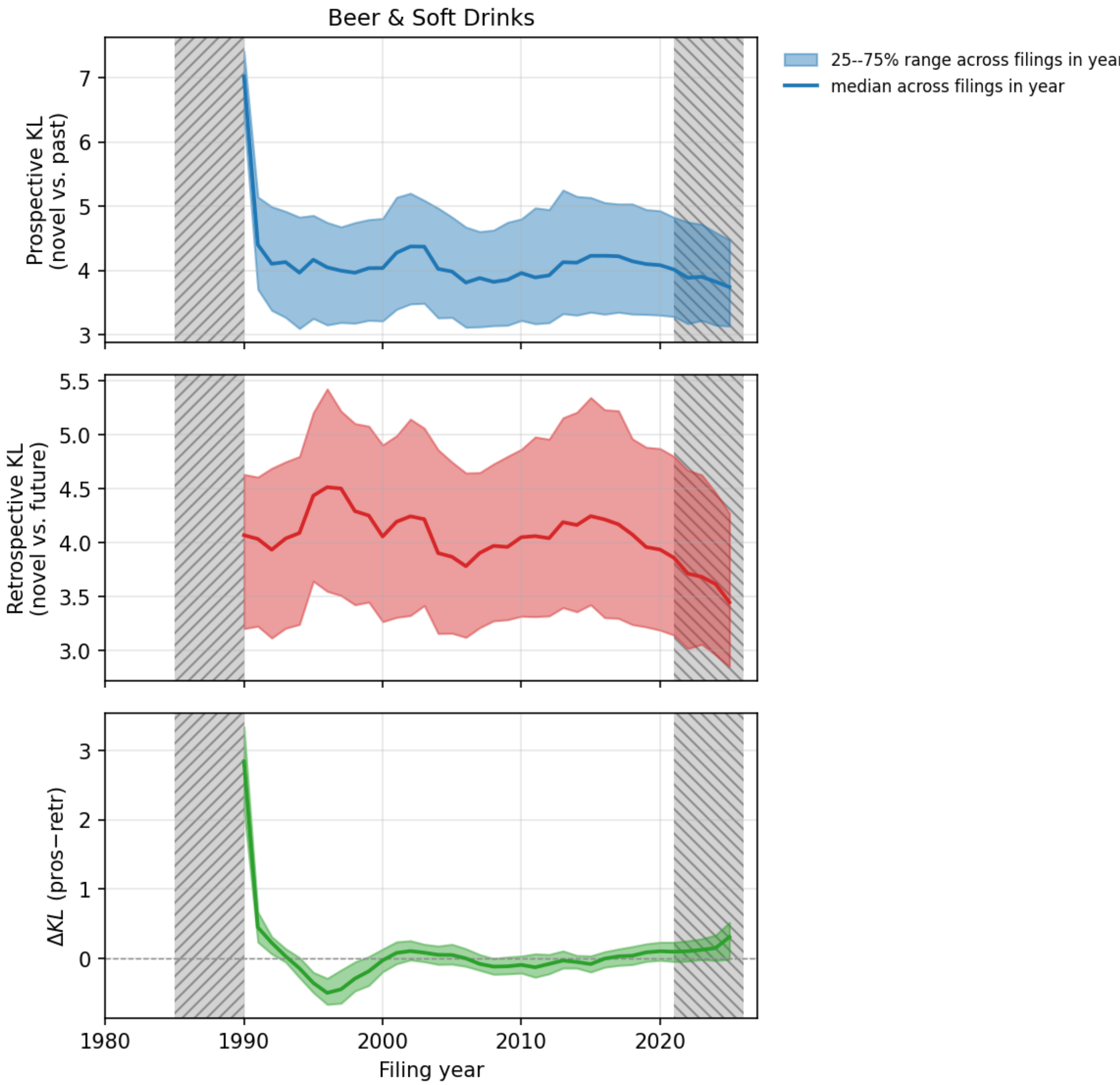

Figure 4: Beer & Soft Drinks: per-year median (line) and 25–75% range across filings in that year (shaded band) of prospective KL, retrospective KL, and ΔKL. Amber/blue: burn-in windows.

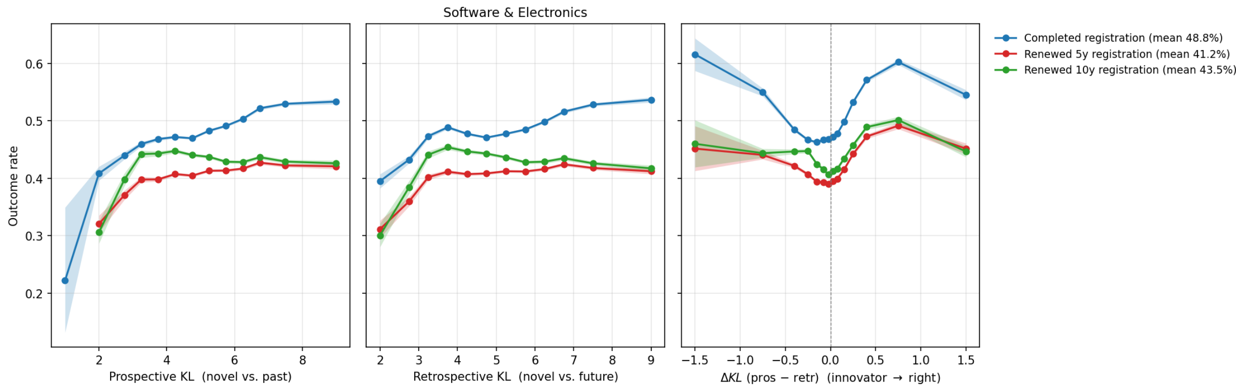

Figure 5: Software & Electronics: registration-and-renewal rates by KL bin. Shaded bands are 95% Wilson confidence intervals on the per-bin mean rate. Pattern across panels: high prospective KL (novelty against the past) penalises both registration and renewals; high retrospective KL (the wording remained unusual in hindsight) is neutral-to-favorable; the difference ΔKL inherits a downward slope.

Figure 6: Beer & Soft Drinks: registration-and-renewal rates by KL bin. Shaded bands: 95% Wilson confidence intervals on the per-bin mean rate.

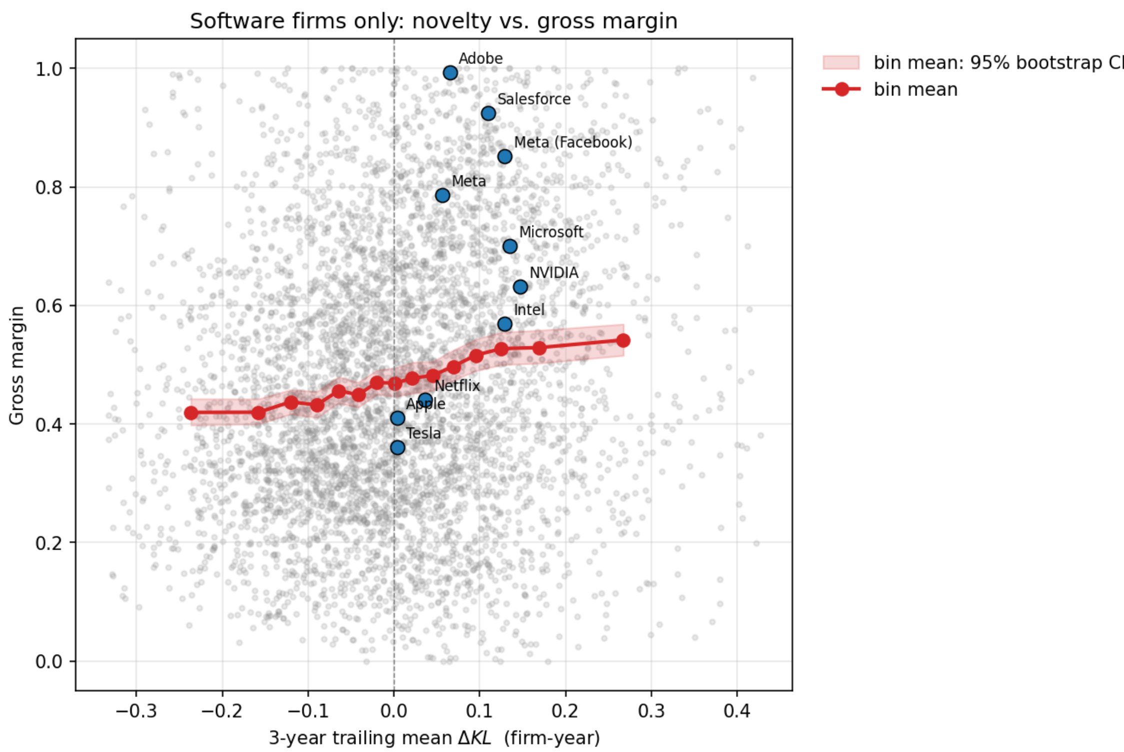

Figure 7: Software-only firm-year panel: gross margin vs. 3-year trailing mean ΔKL. Grey: the underlying 4,044 firm-year observations across 1,366 public software firms (1st–99th percentile of ΔKL, gross margin clamped to $[0, 1]$). Red line: bin means; red shaded band: 95% bootstrap confidence interval on each bin's mean gross margin. Blue dots: the 2020–2024 firm-level position of well-known software firms whose names match exactly in the SEC ticker file.

Table 7: Stock-return regressions on novelty. Each row is a separate OLS regression with year linear and quadratic controls; cluster-robust standard errors at the firm level; outcome winsorised at p1 / p99. Coefficients are in return units per one σ of the predictor.

Outcome	N	β (per σ)	SE
Debut filing only, predictor = ΔKL			
excess vs. S&P 500, 1y	1,446	+0.0363	0.0184
raw 1y return	1,446	+0.0330	0.0186
excess vs. S&P 500, 2y	1,446	+0.0667	0.0272
raw 2y return	1,446	+0.0591	0.0269
excess vs. S&P 500, 3y	1,446	+0.1920	0.0552
raw 3y return	1,446	+0.1732	0.0555
excess vs. S&P 500, 4y	1,447	+0.2844	0.0861
raw 4y return	1,447	+0.2617	0.0872
excess vs. S&P 500, 5y	1,447	+0.1488	0.0652
raw 5y return	1,447	+0.1273	0.0659
Debut filing only, predictor = Prospective KL			
excess vs. S&P 500, 1y	1,446	+0.0008	0.0205
raw 1y return	1,446	+0.0019	0.0208
excess vs. S&P 500, 2y	1,446	-0.0048	0.0258
raw 2y return	1,446	-0.0092	0.0260
excess vs. S&P 500, 3y	1,446	-0.0144	0.0385
raw 3y return	1,446	-0.0166	0.0385
excess vs. S&P 500, 4y	1,447	-0.0181	0.0537
raw 4y return	1,447	-0.0198	0.0539
excess vs. S&P 500, 5y	1,447	+0.0572	0.0521
raw 5y return	1,447	+0.0554	0.0522
Firm-year mean, predictor = ΔKL			
excess vs. S&P 500, 1y	16,334	+0.0151	0.0051
raw 1y return	16,334	+0.0118	0.0053
excess vs. S&P 500, 2y	16,332	+0.0355	0.0092
raw 2y return	16,332	+0.0275	0.0095
excess vs. S&P 500, 3y	16,330	+0.0499	0.0134
raw 3y return	16,330	+0.0382	0.0140
excess vs. S&P 500, 4y	16,329	+0.0410	0.0164
raw 4y return	16,329	+0.0296	0.0167
excess vs. S&P 500, 5y	16,329	+0.0159	0.0203
raw 5y return	16,329	+0.0061	0.0205
Firm-year mean, predictor = Prospective KL			
excess vs. S&P 500, 1y	16,334	+0.0010	0.0045
raw 1y return	16,334	+0.0001	0.0046
excess vs. S&P 500, 2y	16,332	-0.0006	0.0079
raw 2y return	16,332	-0.0018	0.0081
excess vs. S&P 500, 3y	16,330	-0.0011	0.0107
raw 3y return	16,330	-0.0020	0.0109
excess vs. S&P 500, 4y	16,329	-0.0064	0.0149
raw 4y return	16,329	-0.0081	0.0150
excess vs. S&P 500, 5y	16,329	-0.0009	0.0194
raw 5y return	16,329	-0.0046	0.0194

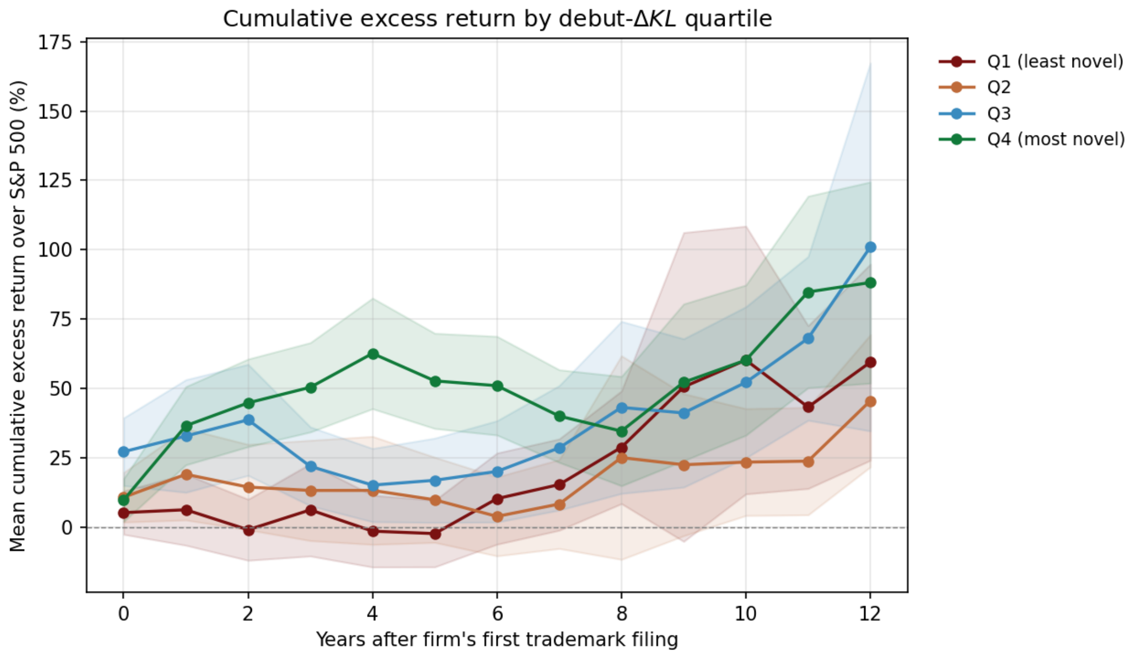

Figure 8: Mean cumulative excess return over the S&P 500 by years after the firm's first trademark filing, split by quartile of debut-filing ΔKL. Shaded bands are 95% standard-error ribbons; only year-since-debut bins with at least 50 firm-years are plotted. The most-novel-debut quartile (Q4, green) accumulates a substantial excess return relative to the median; the least-novel-debut quartile (Q1, dark red) underperforms the benchmark by year three.

Table 8: Per-industry Pearson correlation between firm-mean ΔKL and firm-mean annual excess return over the S&P 500. Asterisk marks correlations with $p < 0.05$. Industries are sorted by matched-firm count.

Industry (NICE)	N firms	r (ΔKL, mean excess)	p	r (pros KL, mean excess)	p
Scientific & Tech Services (042)	1,386	-0.057*	0.035	-0.063*	0.019
Software & Electronics (009)	1,146	-0.013	0.651	+0.047	0.114
Advertising & Retail (035)	1,098	+0.000	0.995	-0.018	0.556
Insurance & Financial (036)	741	+0.071	0.054	+0.145*	0.000
Education & Entertainment (041)	735	+0.028	0.441	+0.064	0.083
Pharmaceuticals (005)	604	+0.026	0.521	+0.066	0.107
Paper, Print & Office (016)	522	+0.050	0.251	+0.006	0.884
Construction & Repair (037)	518	+0.026	0.551	+0.021	0.632
Materials Treatment (040)	440	-0.036	0.457	+0.019	0.690
Telecommunications (038)	395	-0.066	0.191	+0.072	0.154
Medical & Personal Care (044)	383	-0.041	0.426	-0.059	0.247
Transport & Logistics (039)	379	+0.006	0.902	-0.094	0.066
Chemicals & Industrial (001)	346	-0.062	0.251	+0.117*	0.029
Medical Devices (010)	327	+0.069	0.211	+0.021	0.706
Clothing, Footwear & Headwear (025)	320	-0.014	0.799	+0.034	0.548
Legal & Personal Services (045)	319	-0.036	0.524	-0.023	0.686
Machines & Power Tools (007)	314	-0.037	0.512	+0.068	0.231
Lighting, HVAC & Appliances (011)	282	-0.032	0.596	+0.111	0.062
Common Metals (006)	275	+0.002	0.973	-0.010	0.875
Furniture (020)	247	-0.015	0.820	+0.101	0.114

Table 9: Excess-return coefficients with and without the mega-cap cohort (Apple, Microsoft, Amazon, Meta, Alphabet, Netflix, NVIDIA, Tesla). Each cell is one OLS regression; coefficients are excess return per σ of ΔKL, year-and-year-squared controls, cluster-robust SEs.

Horizon	Firm-year (full)	Firm-year (excl. mega-cap)	Debut (full)	Debut (excl. mega-cap)
1y	+0.0151 (0.0051)	+0.0141 (0.0051)	+0.0363 (0.0184)	+0.0362 (0.0184)
2y	+0.0355 (0.0092)	+0.0320 (0.0091)	+0.0667 (0.0272)	+0.0665 (0.0273)
3y	+0.0499 (0.0134)	+0.0444 (0.0131)	+0.1920 (0.0552)	+0.1908 (0.0553)
4y	+0.0410 (0.0164)	+0.0345 (0.0161)	+0.2844 (0.0861)	+0.2823 (0.0866)
5y	+0.0159 (0.0203)	+0.0045 (0.0195)	+0.1488 (0.0652)	+0.1486 (0.0654)

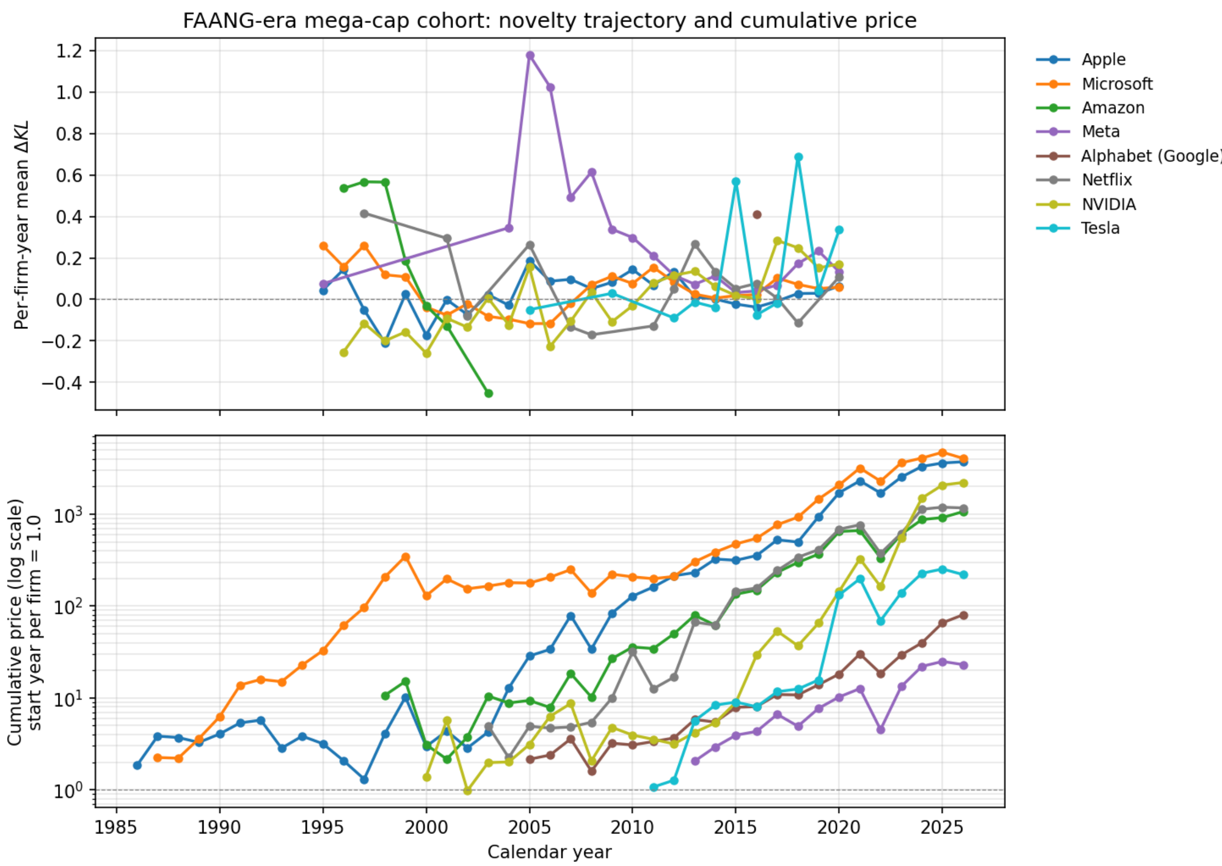

Figure 9: Per-year mean ΔKL (top) and cumulative end-of-year price normalised to first observation (bottom, log scale) for the mega-cap cohort. Tesla's trademark portfolio shows the highest per-year ΔKL in the cohort, consistent with both its linguistic novelty and its outsized stock appreciation; NVIDIA shows elevated ΔKL in the AI era. Apple's late-period ΔKL is comparatively flat (its categories are well-established) yet its cumulative return is among the highest — a reminder that this measure does not capture every channel through which firms create shareholder value.

Table 10: Convergent-validity correlations between trademark ΔKL and patent counts on the matched panel. Column "N" is firm-years (top two rows) or firms (bottom two rows).

Aggregation	N	Pearson r
Firm-year, n_{patents}	174,569	+0.010
Firm-year, $\log(1 + n_{\text{patents}})$	174,569	-0.009
Firm total, sum patents	14,463	+0.008
Firm total, $\log(1 + \text{sum patents})$	14,463	-0.023

Table 11: Excess-return regression with $\log(1 + n_{\text{patents}})$ as the firm-year predictor (z-scored). Same panel and specification as the firm-year row of Table 7.

Horizon	N	β for $\log(1 + \text{patents})_z$	SE
1y	19,686	+0.0203	0.0047
2y	18,553	+0.0429	0.0095
3y	17,394	+0.0591	0.0142
4y	16,245	+0.0902	0.0201
5y	15,141	+0.1206	0.0276

Table 12: Cumulative R^2 from each combination of predictors on excess return at horizon k, on the common panel of firm-years with both a patent count and a trademark KL score. All predictors z-scored; year linear and quadratic controls in every cell; cluster-robust SEs at the firm level (suppressed). The marginal contribution of each predictor is the difference from the controls-only baseline.

Specification (predictors, all z-scored)	1y R^2	2y R^2	3y R^2	4y R^2	5y R^2
Controls only	0.0056	0.0081	0.0129	0.0211	0.0276
Patents (P)	0.0062	0.0089	0.0145	0.0235	0.0306
Innovator dKL (I)	0.0075	0.0111	0.0161	0.0257	0.0312
Novelty pros (S)	0.0056	0.0081	0.0130	0.0212	0.0276
Patents + Innovator (P+I)	0.0081	0.0119	0.0176	0.0280	0.0342
Patents + Novelty (P+S)	0.0062	0.0090	0.0147	0.0236	0.0308
Innovator + Novelty (I+S)	0.0076	0.0111	0.0161	0.0257	0.0312
All three (P+I+S)	0.0082	0.0119	0.0177	0.0280	0.0342
N (firm-years)	8,118	8,116	8,114	8,114	8,112

Table 13: Trademark ΔKL versus four external public innovation comparators. R&D-to-revenue and Crunchbase total funding are continuous, so we report Pearson r. BCG Most Innovative Companies and MIT Technology Review 50 Smartest Companies are binary list memberships, so we report the difference in mean firm ΔKL between firms on the list and the rest of the matched universe (positive = listed firms are more innovative by our measure).

Comparator	Statistic	N matched	Value
R&D-to-revenue (firm-mean)	Pearson r	1,593	+0.0458
BCG Most Innovative (2021–23, union)	On-list vs off-list $\Delta\overline{\Delta KL}$	31	+0.0277
MIT TR 50 Smartest (2015)	On-list vs off-list $\Delta\overline{\Delta KL}$	17	+0.0648
Crunchbase total funding (log)	Pearson r	1,654	-0.0162

Table 14: Pooled quadratic logistic regression of survival outcomes on z-scored ΔKL and its square. Year and year-squared controls suppressed. A positive quadratic coefficient confirms concave-up ("U-shape").

Outcome	N	Linear $\beta^z_{\Delta KL}$	Quadratic $\beta^z_{(\Delta KL)^2}$
Completed registration	9,255,511	-0.0258 (0.0009)	+0.0053 (0.0002)
Renewed 5y registration	9,255,511	-0.0133 (0.0009)	+0.0002 (0.0002)
Renewed 10y registration	6,362,813	+0.0078 (0.0011)	+0.0013 (0.0002)

Table 15: Per-industry quadratic coefficient on $z(\Delta KL)^2$ for 5y registration-renewal. Top 15 (most pronounced U) and bottom 5. Asterisk = $|t| > 1.96$.

Industry (NICE)	N	Linear $\beta^z_{\Delta KL}$	Quadratic $\beta^z_{(\Delta KL)^2}$
Legal & Personal Services (045)	114,693	-0.0256	+0.0307*
Household Utensils (021)	183,800	-0.1586	+0.0258*
Materials Treatment (040)	87,017	+0.0204	+0.0234*
Furniture (020)	144,864	-0.1177	+0.0200*
Common Metals (006)	89,730	-0.1260	+0.0188*
Textiles & Bedding (024)	84,770	-0.1196	+0.0183*
Medical Devices (010)	174,201	-0.0694	+0.0183*
Ropes, Tents & Textiles (raw) (022)	21,551	-0.0989	+0.0183*
Rubber & Insulators (017)	48,186	-0.0901	+0.0169*
Restaurants & Hotels (043)	122,371	-0.0151	+0.0166*
Medical & Personal Care (044)	167,607	+0.0084	+0.0158*
Firearms & Pyrotechnics (013)	12,432	-0.0171	+0.0141*
Leather & Luggage (018)	133,408	-0.1136	+0.0136*
Hand Tools (008)	62,591	-0.0693	+0.0115*
Toys, Games & Sporting Goods (028)	292,436	-0.0697	+0.0105*
...bottom 5 (most negative quadratic = inverted-U):			
Telecommunications (038)	162,634	-0.0261	-0.0130*
Beer & Soft Drinks (032)	94,920	+0.0506	-0.0148*
Agriculture & Live Animals (031)	59,590	+0.0791	-0.0169*
Carpets & Wall Hangings (027)	22,964	-0.0152	-0.0232*
Musical Instruments (015)	11,068	-0.0761	-0.0525*

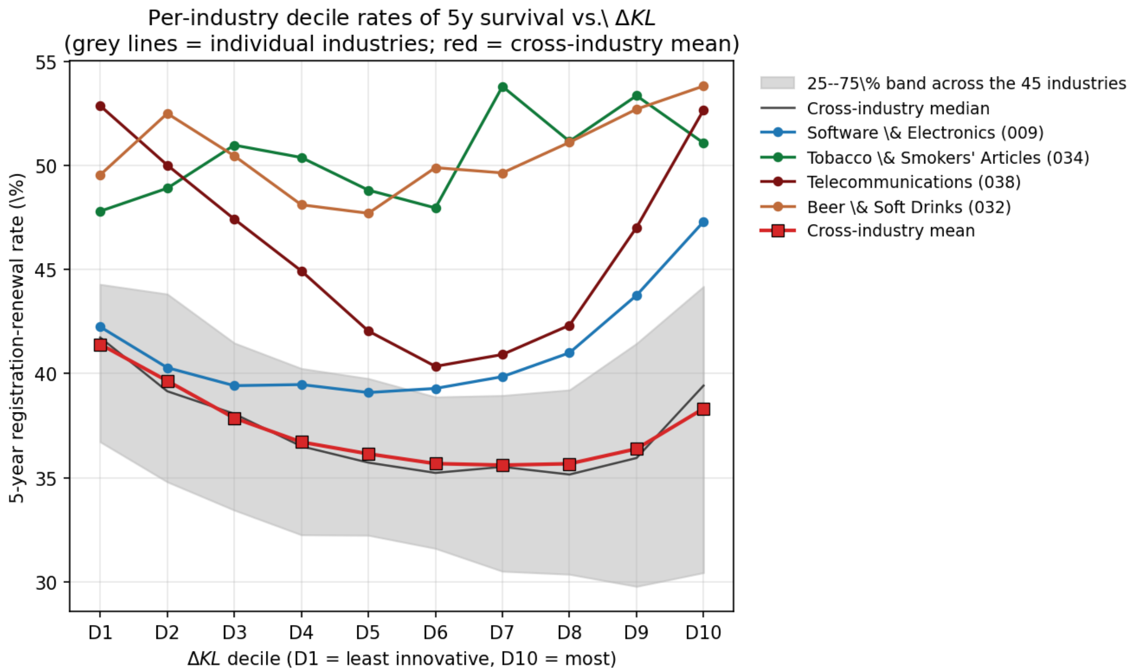

Figure 10: Per-decile 5y registration-renewal rate by ΔKL decile. Each grey line is one of the 45 NICE industries; red is the cross-industry mean. The U-shape is visible in nearly every industry.

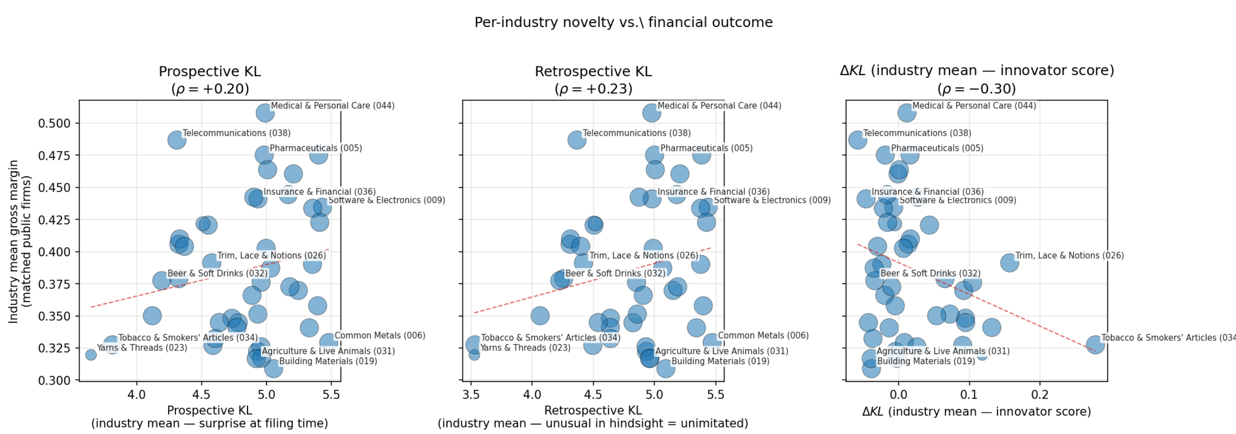

Figure 11: Each dot is one of the 45 NICE international classes; dot size is proportional to the number of matched public firms in the class. Left panel: industry-mean prospective KL (surprise at filing time) on the x-axis. Centre: industry-mean retrospective KL (remained-novel-in-hindsight). Right: industry-mean ΔKL (imitated-innovator score). All three are plotted against industry-mean gross margin on the y-axis. Dashed lines are OLS fits, ρ is the Pearson correlation across the 45 industries.

Table 16: Per-industry quadratic on 5y-survival, sorted by t on the quadratic coefficient (positive = U-shape, negative = inverted-U).

Industry (NICE)	N	β^z_{dKL} (linear)	$\beta^z_{dKL^2}$ (quadratic)	t on quadratic
Industries with the strongest U-shape (top 15 by t on β_{dKL^2})				
Household Utensils (021)	183,800	-0.1586	+0.0258	+14.6
Legal & Personal Services (045)	114,693	-0.0256	+0.0307	+14.5
Materials Treatment (040)	87,017	+0.0204	+0.0234	+12.3
Medical Devices (010)	174,201	-0.0694	+0.0183	+10.9
Restaurants & Hotels (043)	122,371	-0.0151	+0.0166	+9.7
Medical & Personal Care (044)	167,607	+0.0084	+0.0158	+9.4
Furniture (020)	144,864	-0.1177	+0.0200	+9.4
Common Metals (006)	89,730	-0.1260	+0.0188	+8.7
Leather & Luggage (018)	133,408	-0.1136	+0.0136	+8.5
Toys, Games & Sporting Goods (028)	292,436	-0.0697	+0.0105	+8.2
Textiles & Bedding (024)	84,770	-0.1196	+0.0183	+7.8
Cosmetics & Cleaning (003)	300,326	-0.0388	+0.0074	+7.0
Lighting, HVAC & Appliances (011)	171,839	-0.0528	+0.0105	+6.2
Rubber & Insulators (017)	48,186	-0.0901	+0.0169	+5.3
Chemicals & Industrial (001)	118,548	+0.0147	+0.0094	+4.4
Industries with the weakest or inverted U (bottom 5)				
Telecommunications (038)	162,634	-0.0261	-0.0130	-10.6
Advertising & Retail (035)	924,501	+0.0143	-0.0048	-10.7
Scientific & Tech Services (042)	692,938	+0.0168	-0.0097	-12.9
Insurance & Financial (036)	374,634	+0.0541	-0.0122	-13.8
Software & Electronics (009)	1,146,078	+0.0715	-0.0107	-18.2

Table 17: Pooled quadratic logit of 5y survival on $z(\Delta KL)$ and $z(\Delta KL)^2$, with year and year-squared controls, re-fit after dropping different parts of the sample. The quadratic coefficient on ΔKL^2 is essentially unchanged across drop choices — the U-shape is a property of the central distribution, not of the extreme tails.

Sample	N	β^z_{dKL}	$\beta^z_{dKL^2}$	t on quad.
Full sample (none dropped)	9,258,400	-0.0134	+0.0002	+1.3
Drop bottom 5% of prospective KL (¡ 3.05)	8,795,480	-0.0067	-0.0003	-1.7
Drop bottom 5% of retrospective KL (¡ 3.05)	8,795,480	-0.0090	-0.0002	-1.0
Drop bottom 5% of both axes	8,726,416	-0.0058	-0.0007	-3.4
Drop bottom + top 10% of ΔKL (test extreme tails)	7,406,720	-0.0188	+0.0343	+51.2

Table 18: Robustness of the survival U-shape to five candidate confounds. The quadratic coefficient $\beta^z_{\Delta KL^2}$ stays strongly positive everywhere except in the financial outcome, where the relationship is roughly linear.

Sub-sample	N	Outcome rate	Linear $\beta^z_{\Delta KL}$	Quadratic $\beta^z_{\Delta KL^2}$
Foreign-extension hypothesis				
Full sample	600,000	39.2%	+0.0043 (0.0028)	+0.0089 (0.0008)
US-domiciled only (foreign proxy = False)	577,392	39.9%	+0.0043 (0.0028)	+0.0078 (0.0008)
Foreign-domiciled only (foreign proxy = True)	22,608	21.3%	+0.0741 (0.0165)	+0.0100 (0.0038)
First-time vs recurring				
First-time filings only	429,815	39.0%	+0.0019 (0.0033)	+0.0080 (0.0009)
Recurring (2nd+) filings only	170,185	39.6%	+0.0084 (0.0051)	+0.0140 (0.0016)
Financial U-shape: firm-year gross margin				
Public firms (gross margin OLS)	8,600	—	+0.0020 (0.0047)	-0.0022 (0.0012)

Table 19: Single-predictor regression on $|\Delta KL|$ (the absolute distance from $\Delta KL = 0$).

Outcome	N	$\beta^z_{\lvert\Delta KL\rvert}$ (logit)	z	p	McFadden pseudo-R^2	ΔR^2 (linear, vs. controls)
Completed registration	500,000	+0.0022 (0.0031)	+0.7	4.8e-01	0.0246	0.000 pp
Renewed 5y	500,000	+0.0004 (0.0031)	+0.1	8.9e-01	0.0059	0.000 pp
Renewed 10y	343,619	+0.0256 (0.0034)	+7.5	6.0e-14	0.0086	0.016 pp

Table 20: Per-industry summary statistics, restricted to filings from 1995–2020 with both 5-year reference windows populated.

Industry (NICE)	Filings	Clean	Owners	Years	ΔKL med.	ΔKL mean	5y reg. rate	Reg. rate	Top 3 firms by firm-mean ΔKL ($\geq$ 3 filings)
Tobacco & Smokers' Articles (034)	82,520	32,818	16,646	1990–2020	+0.15	+0.29	50%	56%	PHILLIPS & KING INTERNATIONAL, INC. (+3.12); AUDIO TECHNOLOGY OF NEW YORK INC. (+2.75); BAMBU SALES, INC. (+2.68)
Trim, Lace & Notions (026)	84,034	35,638	24,532	1990–2020	+0.10	+0.17	36%	43%	Selected Trading Corp. (+2.49); LITTLE ROSEY ENTERPRISES, INC. (+2.19); XUCHANG BEST TRADING CO.,LTD (+1.93)
Ropes, Tents & Textiles (raw) (022)	47,287	21,551	14,335	1990–2020	+0.08	+0.16	31%	42%	International Hairgoods, Inc. (+2.10); TENSILE SHADE PRODUCTS, LLC (+1.96); Hoechst Celanese Corporation (+1.88)
Household Utensils (021)	403,933	183,800	114,891	1990–2020	+0.06	+0.12	36%	43%	BARRON, THOMAS ARCHIBALD (+2.90); Franzus Company, Inc. (+2.82); Villeroy & Boch S.a.r.l (+2.60)
Yarns & Threads (023)	15,210	2,889	1,756	2006–2020	+0.10	+0.12	24%	31%	QINGDAO BANGTE FIBRE CO., LTD. (+0.82); Champion Thread Company (+0.72); Jiangsu Lugang Science & Technology Co., (+0.65)
Hand Tools (008)	128,745	62,591	38,099	1990–2020	+0.06	+0.12	32%	42%	EDDIE BAUER, INC. (+2.36); O.M.S. Inc. (+2.26); WASHINGTON FORGE, INCORPORATED (+2.08)
Carpets & Wall Hangings (027)	56,986	22,964	12,990	1990–2020	+0.07	+0.12	35%	43%	Hallmark Cards, Incorporated (+2.31); Armstrong World Industries, Inc. (+1.85); Kentile Floors Inc. (+1.85)
Furniture (020)	326,621	144,864	87,163	1990–2020	+0.04	+0.11	36%	44%	Express Yourself, Inc. (+3.02); Helikon Furniture Company, Inc. (+2.93); Ohio Mattress Company Licensing And Comp (+2.70)
Lighting, HVAC & Appliances (011)	333,507	171,839	95,942	1990–2020	+0.04	+0.09	31%	41%	Hiross, Inc. (+2.79); Melitta-Werke Bentz & Sohn (+2.66); Water Resources International Incorporat (+2.63)
Textiles & Bedding (024)	173,759	84,770	50,657	1990–2020	+0.05	+0.08	38%	45%	FORBES INC. (+2.55); AD PROPERTIES CORP. (+2.47); Hallmark Cards, Incorporated (+2.46)
Firearms & Pyrotechnics (013)	37,403	12,432	6,331	1992–2020	+0.07	+0.08	33%	47%	Wright Leather Works LLC (+1.52); G.A. Precision LLC (+1.43); MAGPUL INDUSTRIES CORP. (+1.13)
Wine & Spirits (033)	222,138	61,865	29,925	1990–2020	+0.03	+0.07	46%	53%	Two Beers Brewing Company (+2.60); SNOW BRAND MILK PRODUCTS CO., LTD. (+2.50); Bodegas "El Pomar" S. A. de C.V. (+2.48)
Vehicles (012)	242,485	116,643	61,001	1990–2020	+0.01	+0.06	34%	45%	MIYATA INDUSTRY CO., LTD. (+3.53); Marui Co., Ltd. (+3.52); JENSEN CAR COMPANY LIMITED (+3.28)
Jewelry & Watches (014)	234,908	98,618	64,105	1990–2020	+0.02	+0.06	38%	45%	BURRELL, KIRK (+2.63); IRENEW BIO ENERGY SOLUTIONS LLC (+2.51); National Trading Mfg., Inc. (+2.51)
Medical Devices (010)	302,192	174,201	79,685	1990–2020	-0.01	+0.05	38%	48%	Mikrona Technologie AG (+2.86); AudioScience, Inc. (+2.85); Inventions Extraordinaire Inc. (+2.82)
Toys, Games & Sporting Goods (028)	583,108	292,436	144,975	1990–2020	-0.01	+0.04	42%	49%	American Video Entertainment, Inc. (+3.21); VIRUSTATIC INC. (+3.07); Abu Aktiebolag (+2.93)
Cosmetics & Cleaning (003)	564,014	300,326	139,169	1990–2020	+0.02	+0.04	43%	51%	MARKETING SERVICES OF BEND, LLC (+3.64); Image Laboratories, Inc. (+3.06); PALEO PAW CORP. (+2.95)
Common Metals (006)	175,437	89,730	51,240	1990–2020	+0.01	+0.04	29%	45%	Shell Oil Company (+3.03); SOMBUR MACHINE AND TOOL (+2.64); LANAI RESORT PARTNERS (+2.47)
Leather & Luggage (018)	249,052	133,408	84,643	1990–2020	-0.01	+0.04	38%	45%	Tonka Corporation (+2.97); Grifs Western, Inc. (+2.72); Impuls International, Inc. (+2.62)
Rubber & Insulators (017)	88,430	48,186	24,876	1990–2020	+0.00	+0.04	28%	44%	Filmco Industries, Inc. (+2.31); Oregon Research and Development Corporat (+1.97); Stone Container Corporation (+1.86)
Lubricants & Fuels (004)	78,345	32,824	17,794	1990–2020	-0.02	+0.03	38%	49%	Ashland Oil, Inc. (+2.66); Geo. Pfau's Sons Company, Inc. (+1.92); TOPCO ASSOCIATES, INC. (+1.62)
Restaurants & Hotels (043)	285,610	122,371	74,815	2003–2020	+0.02	+0.03	39%	49%	Burger Bench, LLC (+2.82); RIB CITY GROUP, LLC (+2.32); Pizza Joints Inc. (+1.83)
Software & Electronics (009)	1,807,636	1,146,078	558,481	1990–2020	-0.02	+0.02	41%	49%	Til Systems Limited (+3.60); Northgate Computer Systems, Inc. (+3.30); JAMES RIVER II, INC. (+3.24)
Machines & Power Tools (007)	286,930	148,670	76,188	1990–2020	-0.02	+0.02	27%	41%	Republic Machinery Co., Inc. (+2.99); Rietschle, Inc. (+2.64); NLB CORP. (+2.59)
Coffee, Tea, Bakery & Spices (030)	428,854	192,476	97,520	1990–2020	-0.01	+0.02	43%	52%	GMB Enterprises, Inc. (+3.24); Buy The Weigh, Inc. (+3.07); Cafetalera Zardain Hermanos S.A. de C.V. (+2.96)
Paper, Print & Office (016)	712,163	419,972	229,601	1990–2020	-0.01	+0.02	41%	51%	LawPrep, Inc. (+2.84); Anderson Import-Export Company, Inc. (+2.79); CLASSIC GAMES, INC. (+2.77)
Education & Entertainment (041)	1,344,519	850,779	491,343	1990–2020	-0.01	+0.02	38%	48%	Corporacion Wilfrido Vargas, S.A. (+3.98); CENTURY OCN PROGRAMMING, INC. (+3.86); Video Challenge, Inc. (+3.67)
Materials Treatment (040)	145,590	87,017	55,201	1990–2020	-0.01	+0.02	33%	45%	Re-USA-ble, Inc. (+3.33); Lynn, Robert C (+3.05); SUNMARKS, INC. (+3.00)
Chemicals & Industrial (001)	218,420	118,548	49,757	1990–2020	-0.01	+0.02	33%	48%	Seattle FilmWorks, Inc. (+3.07); Artis, Inc. (+2.86); International Flavors & Fragrances, Inc. (+2.57)
Construction & Repair (037)	284,361	166,789	99,723	1990–2020	-0.00	+0.01	30%	43%	VERITECHNOLOGY ELECTRONICS CORPORATION (+3.12); All State Plumbing Inc. (+2.73); American Lubrication Services, Inc. (+2.69)
Agriculture & Live Animals (031)	136,656	59,590	30,003	1990–2020	-0.03	+0.01	40%	50%	Oxnard Vegetable Exchange, Inc. (+3.37); CENTRAL GARDEN SUPPLY (+3.37); PET-AG, INC. (+3.32)
Medical & Personal Care (044)	312,270	167,607	97,110	2003–2020	-0.00	+0.01	37%	46%	Golden, Joseph (+2.16); Logue, Charles G. (+2.16); My Therapy Session, Inc. (+1.87)
Pharmaceuticals (005)	600,473	336,093	124,260	1990–2020	-0.01	+0.01	49%	57%	Marlyn Company, Inc. (+4.80); WINNING LAB. INC. (+4.46); P. Leiner Nutritional Products, Inc. (+4.34)
Scientific & Tech Services (042)	1,148,416	692,938	387,431	1990–2020	+0.01	+0.01	39%	47%	SOUTHWEST TEXAS METHODIST HOSPITAL (+3.54); Treehouses of America, Inc. (+3.48); Premier Cruise Lines, Ltd. (+3.37)
Beer & Soft Drinks (032)	206,304	94,920	46,682	1990–2020	-0.02	+0.01	51%	57%	BONFARE FOOD STORES, INC. (+4.24); WILLINGER, DOUGLAS A. (+4.24); Kola Colombiana Inc. (+4.20)
Meat, Dairy & Prepared Foods (029)	253,936	127,008	60,556	1990–2020	-0.03	+0.01	43%	52%	Mrs. Kinney's Kitchen, Inc. (+2.98); Seven Seas Foods, Inc. (+2.98); Curtice-Burns Foods, Inc. (+2.94)
Building Materials (019)	122,596	66,282	33,012	1990–2020	-0.02	+0.00	32%	46%	Manville Corporation (+2.86); FLORIDA TILE INDUSTRIES, INC. (+2.65); SENECA TILES, INC. (+2.65)
Paints & Coatings (002)	64,892	31,911	15,703	1990–2020	-0.04	+0.00	30%	46%	Tintas Coral S/A (+3.15); ABC Protective Coating Corporation (+2.56); Graphic Color Corporation (+2.49)
Legal & Personal Services (045)	203,476	114,693	74,406	2003–2020	-0.01	+0.00	36%	46%	Sproose, Inc. (+1.96); Katz, Daniel (+1.81); SkyWard IO, Inc. (+1.74)
Transport & Logistics (039)	187,535	112,578	63,048	1990–2020	-0.01	+0.00	34%	46%	Alamo Rent-A-Car (+3.42); Novair International Airways Ltd. (+3.03); First Family of Travel, Ltd. (+2.88)
Advertising & Retail (035)	1,418,345	924,501	542,287	1990–2020	-0.03	-0.00	38%	47%	TRATEC OF CALIFORNIA, INC. (+3.77); Pioneer TeleTechnologies, Inc. (+3.65); American Family Publishers (+3.65)
Clothing, Footwear & Headwear (025)	1,265,966	604,850	382,965	1990–2020	-0.05	-0.01	47%	53%	Pacific Coast Management Team (+3.38); Ashley Productions, Inc. (+3.35); Carter, Gregory, C (+3.35)
Musical Instruments (015)	31,792	11,068	6,747	1993–2020	-0.00	-0.01	31%	45%	Original Fuzz, LLC (+1.43); Sun Jianjun (+0.85); SUNLAND INFORMATION TECHNOLOGY CO., LTD. (+0.82)
Insurance & Financial (036)	602,901	374,634	183,490	1990–2020	-0.05	-0.02	36%	48%	McDEVITT, MICHAEL J. (+3.66); Visconti, James A. (+3.65); Medical Bill Management, Inc. (+3.63)
Telecommunications (038)	230,267	162,634	82,158	1990–2020	-0.03	-0.03	46%	52%	United Artists Holding, Inc. (+4.37); Ingram, William H. (+4.21); Ohringer, Sheldon S. (+3.91)

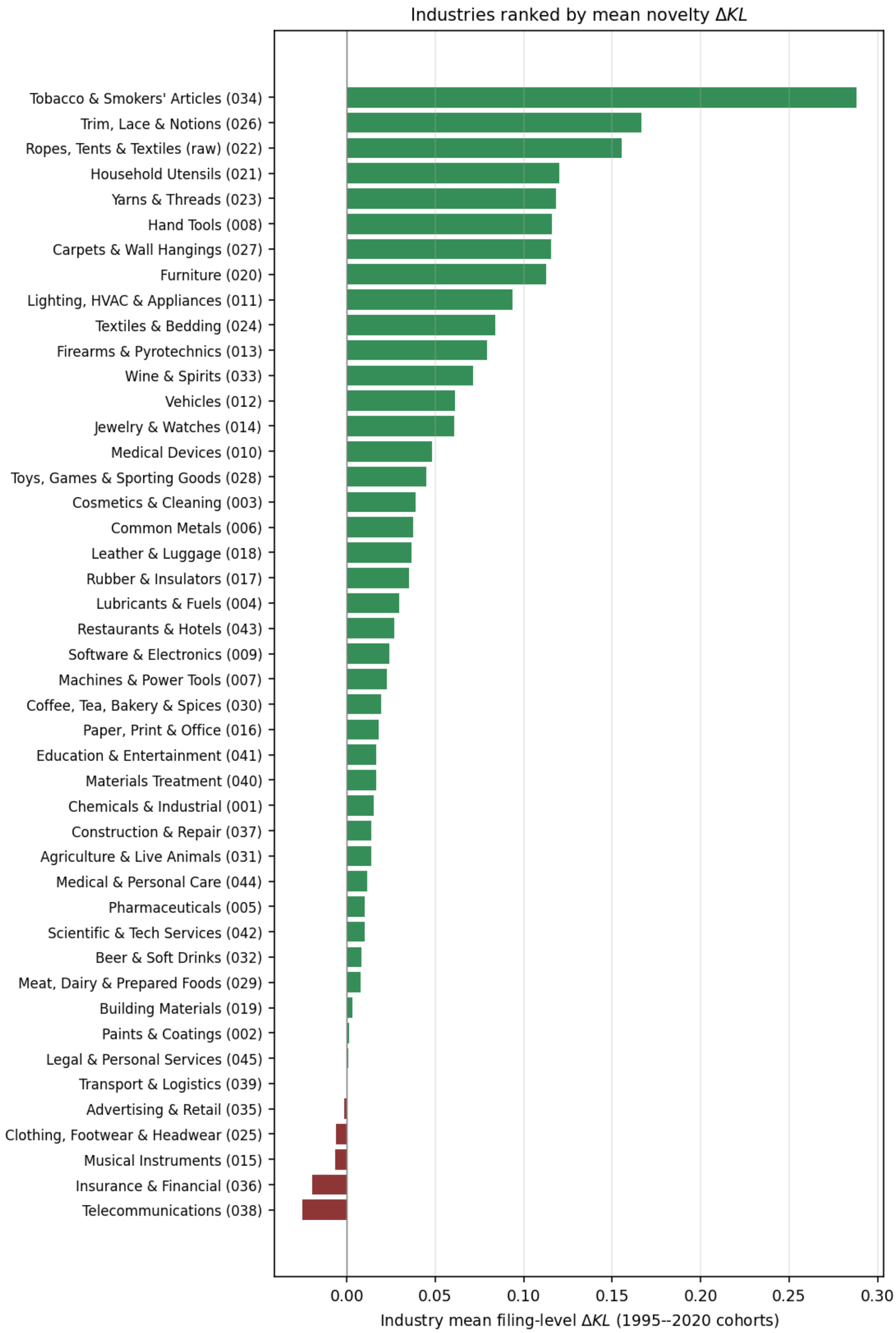

Figure 12: Industries ranked by mean filing-level ΔKL across 1995–2020 cohorts. Green: industries whose typical filing was prospectively novel and retrospectively obvious (the future imitated the past). Red: industries whose typical filing was the opposite. Tobacco & Smokers' Articles tops the list, driven by the 2014–2020 e-cigarette / vape vocabulary explosion. Telecommunications, Insurance & Financial, and Paints & Coatings sit at the bottom: their goods-and-services lexicon was already settled before 1995 and has been stable since.

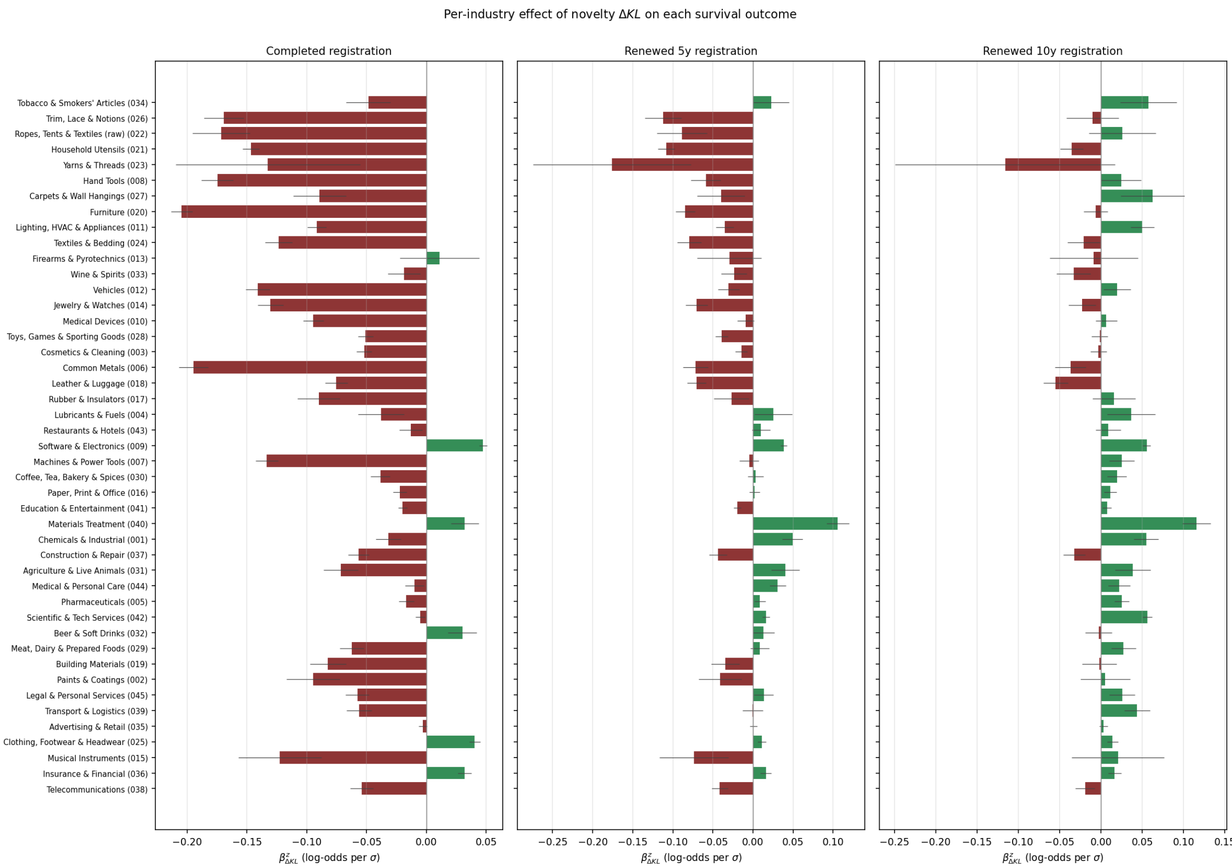

Figure 13: Per-industry effect of ΔKL on survival outcomes (logistic-regression z-score coefficient with 95% CI). Industries are ordered by mean ΔKL (the same order as Figure 12). **The sign of the innovation reward varies by industry.** Some industries (e.g. Common Metals, Hand Tools, Paints & Coatings) actually *reward* novel filings at the registration gate, presumably because in those industries a novel goods/services description can describe a real new use the examiner recognises as patentable activity. Most industries match the software pattern: novelty penalised at registration, neutral or favourable at renewal.

Table 21: Top-1 highest-ΔKL mark per industry, 2010–2020 cohort. The 2014–2020 cannabis/CBD/blockchain wave appears in nearly every consumer-goods class.

Industry (NICE)	Year	ΔKL	Mark	Owner at filing
Tobacco & Smokers' Articles (034)	2016	+4.51	DAISY	Green Tank Technologies Corp.
Agriculture & Live Animals (031)	2018	+4.20	UNITED AMERICAN HEMP	United American Hemp LLC
Cosmetics & Cleaning (003)	2018	+4.05	LEEF	PALEO PAW CORP.
Chemicals & Industrial (001)	2017	+4.03	REEFERTILIZER	Reefertilizer Inc.
Coffee, Tea, Bakery & Spices (030)	2012	+3.93	RED & WHITE	Forefront Enterprises, LLC
Household Utensils (021)	2017	+3.59	IREPELLER	10093939 Canada Corp.
Beer & Soft Drinks (032)	2018	+3.56	KANNABIER	Advanced Creative Intelligent
Wine & Spirits (033)	2014	+3.49	CHATEAU DE CAYX	Boutique du Château de Cayx
Ropes, Tents & Textiles (raw) (022)	2013	+3.33	SUNAMI	TENSILE SHADE PRODUCTS, LLC
Scientific & Tech Services (042)	2014	+3.30	ZENPOOL	Geniuses at Work Corporation
Toys, Games & Sporting Goods (028)	2012	+3.20	BETCLOUD	BALLY GAMING, INC.
Materials Treatment (040)	2018	+3.17	VERITAS — FARMS	271 Lake Davis Holdings LLC
Meat, Dairy & Prepared Foods (029)	2016	+3.15	PHIVIDA	PHIVIDA HOLDINGS INC.
Jewelry & Watches (014)	2013	+3.11	OFF THE CLOSET	Rahhal, Nour
Software & Electronics (009)	2018	+3.09		Unify Earth UEX
Furniture (020)	2016	+3.07	GO PET CLUB	Pel Wholesale, Inc.
Paper, Print & Office (016)	2020	+3.05	KAYLEEMAY LLC	Fall, Cayle M
Clothing, Footwear & Headwear (025)	2010	+3.03	CHARMANT FEMME	2BB UNLIMITED, INC.
Medical Devices (010)	2020	+3.03	ELASTIMASK	Bedford Industries, Inc.
Education & Entertainment (041)	2017	+3.01	ACADEMY	Kingsland School of Blockchain
Trim, Lace & Notions (026)	2015	+2.99	WOWQUEEN	XUE YONG BO
Advertising & Retail (035)	2018	+2.85	OPENSEA	Ozone Networks, Inc.
Lubricants & Fuels (004)	2017	+2.82		Mary Jane's Medicinals, LLC
Pharmaceuticals (005)	2017	+2.74	CANINES BEST DEFENSE CBD	Nobrega, Kathryn
Insurance & Financial (036)	2017	+2.71	MOSAIC TOKEN	MOSAIC RESEARCH LIMITED
Medical & Personal Care (044)	2018	+2.64	FARMING A HEALTHIER FUTURE	Southern Hemp, LLC
Carpets & Wall Hangings (027)	2016	+2.63	LOVEMAT	Wuhan Xuanyuanlong Science and
Transport & Logistics (039)	2017	+2.63	MARIJUANA EXPRESS	Marijuana Express LLC
Lighting, HVAC & Appliances (011)	2010	+2.42	KESSIL	DiCon Fiberoptics, Inc.
Vehicles (012)	2013	+2.34	PRECISION DRONE WWW.PRECISIONDRO	Precision Drone, LLC
Leather & Luggage (018)	2018	+2.31	NOMADZ	Ambit Global
Textiles & Bedding (024)	2017	+2.27	BESTDROP	Zhiyong Xie
Legal & Personal Services (045)	2011	+2.26	DIAMONDS IN THE RUFF	CINDY ROSS
Restaurants & Hotels (043)	2013	+2.20	GREEN FEET BREWING	Green Feet Brewing
Hand Tools (008)	2016	+2.02	THE NAZOR	Nazor, LLC
Firearms & Pyrotechnics (013)	2010	+1.94	ALPHA RAIL	TROY INDUSTRIES, INC.
Machines & Power Tools (007)	2017	+1.86	COUNTERMATE	IMPAK Corporation
Musical Instruments (015)	2012	+1.75	ERGODOCK	Hickey, Bruce
Common Metals (006)	2017	+1.70	UP YOUR CASTER	CASTER CONNECTION, LLC
Yarns & Threads (023)	2011	+1.69	BERROCO NANUK	Berroco, Inc.
Rubber & Insulators (017)	2014	+1.69	B3D	Brandon Bell
Construction & Repair (037)	2010	+1.52	WILKS MASONRY	Wilks Masonry Corporation
Building Materials (019)	2012	+1.51	FREE FORM YOGA FLOOR	Isabelle J. Barger
Telecommunications (038)	2017	+1.44	NCRYPT	NCHAIN HOLDINGS LTD.
Paints & Coatings (002)	2016	+1.32	TRIPLE BEST	Sike Wu

Table 22: Selected filings with per-token KL contributions to prospective and retrospective surprise. **Surprise is an interaction across tokens, not a property of any single word.** A filing's KL score is $\sum_v P(v)\log[P(v)/Q(v)]$, summed across every token v that appears in the filing's goods/services description. The token `software` is by itself common in Class 9, but its *combination* with rare neighbors — `generative artificial`, `conversational agents`, `cryptographic keys` — can give a filing high prospective surprise relative to the look-behind window's mix of tokens. Tokens shown below are the top five by per-token contribution $P(v)\log[P(v)/Q(v)]$ to each direction's KL; common tokens like `software` can appear in the prospective list when their multinomial weight in the filing is much larger than their weight in the look-behind reference.

Mark / Year	Owner at filing	KL_{pros}	KL_{retr}	ΔKL	Top tokens driving prospective surprise	Top tokens driving retrospective surprise
OPENAI (2016)	OpenAI, Inc.	7.65	5.56	+2.09	`software artificial`, `artificial intelligence`, `intelligence`, `artificial`, `software`	`software artificial`, `artificial intelligence`, `artificial`, `intelligence`, `software`
CLAUDE (2023)	Anthropic, PBC	7.12	5.06	+2.06	`model performing`, `document question-answering`, `tasks natural`, `generative text`, `performing generative`	`tasks`, `natural`, `model performing`, `question-answering simulating`, `document question-answering`
INSTAGRAM (2011)	INSTAGRAM, LLC	7.16	6.03	+1.13	`appearance enabling`, `modifying appearance`, `software modifying`, `enabling transmission`, `appearance`	`appearance enabling`, `modifying appearance`, `software modifying`, `enabling transmission`, `appearance`
PHOTOSHOP (1992)	ADOBE INC.	4.81	4.83	-0.01	`manipulating graphic`, `creating manipulating`, `images computer`, `graphic images`, `manipulating`	`manipulating graphic`, `creating manipulating`, `images computer`, `graphic images`, `programs creating`
KUBERNETES (2014)	LF PROJECTS, LLC	4.70	4.62	+0.08	`servers secure`, `secure private`, `private networks`, `networks`, `private`	`servers secure`, `secure private`, `private networks`, `networks`, `private`
WHITE CLAW SELTZER WORKS (2016)	MARK ANTHONY INTERNATIONAL S	9.12	7.35	+1.78	`sugar-based beer`, `sugar-based`, `seltzer`, `beer`	`sugar-based beer`, `sugar-based`, `seltzer`, `beer`
LIQUID DEATH (2023)	Supplying Demand, Inc.	3.97	3.80	+0.17	`drink mixes`, `water nutritional`, `mixes concentrates`, `water sparkling`, `concentrates`	`drink mixes`, `mixes concentrates`, `water nutritional`, `concentrates`, `flavor enhanced`
HARD MTN DEW (2021)	PEPSICO, INC.	4.81	4.96	-0.15	`flavored brewed`, `brewed`, `brewed malt-based`, `malt-based alcoholic`, `nature beer`	`flavored brewed`, `brewed`, `brewed malt`, `nature beer`, `malt-based alcoholic`
ROCKSTAR (2002)	ROCKSTAR ENERGY LLC	4.74	3.92	+0.81	`drinks`, `energy drinks`, `energy`	`drinks`, `energy drinks`, `energy`

Limitations

- **Goods/services text is one of several novelty channels.** Brand novelty (Liquid Death), design novelty (AirPods casing), and pricing novelty are invisible to this corpus. Estimates here should be read as innovation-in-business-description novelty.
- **The crosswalk is exact-match only.** A fuzzy-match second pass and manual review of the top-by-volume USPTO owners would plausibly double or triple the matched panel size and reduce survivorship bias toward firms that did not change their registered name.
- **Survivorship bias in the financial panel.** Firms appear in SEC FSDS only while public. Pre-IPO innovators (Anthropic, OpenAI, Stripe at most timepoints) and dead-but-once-public firms drop out. The financials reward we estimate is conditional on already being a public concern.
- **The reference distributions are smoothed multinomials, not topic models.** An LDA replication is the obvious artifact-check.

Next steps

The cross-industry universal sweep (all 45 NICE classes pooled) would pin down which industries are innovative by this measure and would allow new-class births (cannabis beverages 2018, generative AI tools 2022) to be scored against a cross-industry background. A fuzzy-match crosswalk would expand the financial panel from $\sim$ 1,400 firms toward the $\sim$5,000 ceiling of the SEC EDGAR universe. With either or both extensions in place, the natural follow-up is a *variance* regression: do high-ΔKL firms exhibit higher margin variance, the empirical fingerprint of pioneers either spectacularly succeeding or spectacularly failing?

A Per-class summary statistics

Industry (NICE)	Filings	Clean	Years	Owners	ΔKL med.	p5	p95
Software & Electronics (009)	1,807,636	1,515,300	1958–2025	754,869	-0.02	-0.36	+0.45
Beer & Soft Drinks (032)	206,304	125,512	1980–2025	62,180	+0.01	-0.53	+0.60

B Selected filings, full text

OPENAI (2016) Owner at filing: *OpenAI, Inc.*. $KL_{\text{pros}} = 7.65$, $KL_{\text{retr}} = 5.56$, $\Delta KL = +2.09$.
Goods/services text: *Computer software for artificial intelligence*

CLAUDE (2023) Owner at filing: *Anthropic, PBC.* $KL_{\text{pros}} = 7.12$, $KL_{\text{retr}} = 5.06$, $\Delta KL = +2.06$.
Goods/services text: *Downloadable computer software in the nature of an artificial intelligence model for performing generative text AI tasks and natural language processing AI tasks and for writing content based on a theme, summarizing text, document question-answering and simulating natural conversation; downloadable software in the nature of a downloadable mobile application featuring an artificial intelligence model for performing generative text AI tasks and natural language processing AI tasks and for writing content based on a t...*

INSTAGRAM (2011) Owner at filing: *INSTAGRAM, LLC.* $KL_{\text{pros}} = 7.16$, $KL_{\text{retr}} = 6.03$, $\Delta KL = +1.13$.

Goods/services text: *Downloadable computer software for modifying the appearance and enabling transmission of photographs*

PHOTOSHOP (1992) Owner at filing: *ADOBE INC..* $KL_{\text{pros}} = 4.81$, $KL_{\text{retr}} = 4.83$, $\Delta KL = -0.01$.

Goods/services text: *computer programs for creating and manipulating graphic images on a computer [and manuals for use therewith, sold as a unit]*

KUBERNETES (2014) Owner at filing: *LF PROJECTS, LLC.* $KL_{\text{pros}} = 4.70$, $KL_{\text{retr}} = 4.62$, $\Delta KL = +0.08$.

Goods/services text: *Computer software for accessing, browsing, sharing, defining, maintaining, virtualizing and communicating information over computer networks, servers and secure private networks; computer software for use in creating, managing, controlling, changing, replicating, deploying, naming and linking information over computer networks, servers and secure private networks; computer software for assisting users in navigating through computer networks, servers and secure private networks; computer software for running web app...*

WHITE CLAW SELTZER WORKS (2016) Owner at filing: *MARK ANTHONY INTERNATIONAL SRL.* $KL_{\text{pros}} = 9.12$, $KL_{\text{retr}} = 7.35$, $\Delta KL = +1.78$.

Goods/services text: *Hard seltzer Brewed sugar-based beer*

LIQUID DEATH (2023) Owner at filing: *Supplying Demand, Inc..* $KL_{\text{pros}} = 3.97$, $KL_{\text{retr}} = 3.80$, $\Delta KL = +0.17$.

Goods/services text: *Powdered drink mixes for making flavored water, sports drinks, and energy drinks; powders and concentrates for making flavor enhanced water and sparkling water Powdered nutritional supplement drink mixes and concentrates*

HARD MTN DEW (2021) Owner at filing: *PEPSICO, INC..* $KL_{\text{pros}} = 4.81$, $KL_{\text{retr}} = 4.96$, $\Delta KL = -0.15$.

Goods/services text: *flavored brewed malt-based alcoholic beverages in the nature of beer Alcoholic flavored brewed malt beverages, except beers*

ROCKSTAR (2002) Owner at filing: *ROCKSTAR ENERGY LLC.* $KL_{\text{pros}} = 4.74$, $KL_{\text{retr}} = 3.92$, $\Delta KL = +0.81$.

Goods/services text: *Sports drinks, namely, energy drinks*